

Notes on the “pirate” dedisperser: inpainting

Kendrick Smith

September 7, 2026

Contents

1	1-d Kalman inpainting	2
1.1	Problem statement and notation	2
1.2	Algorithms	3
1.3	Summary of results	5
1.4	A weaker figure of merit?	7
A	1-d Kalman inpainting: proofs and details	8
A.1	More notation, and the constants of the smoother	9
A.2	The exact identities	9
A.3	Upper bounds	10
A.4	Plain inpainting: the lower side	11
A.5	Algorithm C: the lower bound	13
A.6	Algorithm A: the b -term	14
A.7	Overmasking cost	15
A.8	Numerical methods and validation	16
A.9	Remarks: algorithm B, fixed lag, renormalization	17
A.10	Dependence on τ	17
A.11	The windowed figure of merit	18

1 1-d Kalman inpainting

The Kalman detrender of *detrending.pdf*, “Time detrending algorithm 2: Kalman filter” sets its residual to zero at masked samples. Downstream, a masked sample therefore enters the dedispersion sums and the peak-finding convolutions (*dedispersion.pdf*, “Peak-finding”) as a zero rather than as a noise sample, so the variance of a peak-finder output depends on how much of its kernel lies on masked samples, and near masks it can be far from the unmasked value. *Inpainting* attacks this at the source: replace every masked sample by the fitted baseline plus simulated noise, detrend again without a mask, and hand the second-pass residual downstream, so that the peak-finder sees something that looks like detrended noise everywhere.

This section asks how well that works. The data are pure noise and the mask is chosen adversarially: how far can the variance of a peak-finder output stray from its unmasked value? The answer turns on an exact identity (§A.2): the second pass cannot repair any variance the *first* pass removed from valid samples, so every large deviation is a leverage effect of the masked detrender, and for plain inpainting it is always a deficit. Two refinements that enlarge the mask where the leverage is high are then analyzed. §1.1 sets up the problem, §1.2 states the candidate algorithms, §1.3 summarizes every result (saying which are theorems and which are numerical), and Appendix A contains the proofs and details.

1.1 Problem statement and notation

Setting. A one-dimensional timestream $d[t]$, $0 \leq t < T$, with a boolean mask $m[t]$ ($1 = \text{valid}$) and noise variance σ^2 independent of t . Throughout, the data are pure noise, $d[t] \sim \mathcal{N}(0, \sigma^2)$ i.i.d.: there is no baseline to subtract, so everything below is a property of the mask and of the algorithm. Both detrender passes are whole-timestream (“one-shot”) solves, i.e. the global penalized least-squares estimator of *detrending.pdf*, “The model”, not the fixed-lag estimator that production uses (§A.9 says what the fixed lag changes), and only samples far from the ends of the timestream are considered, $\min(t, T-t) \gg \tau$. The estimator does not depend on σ , so we set $\sigma = 1$.

The Kalman detrender. The reader is assumed to know *detrending.pdf*, “Time detrending algorithm 2: Kalman filter”; the objects used here are

$$\begin{aligned}
k &= \text{order of the integrated random walk } (k = 2 \text{ throughout}) \\
\tau &= \text{smoothing timescale in samples } (\tau = 64 \text{ throughout; §A.10 for other values}) \\
\rho &= \tau^{2k} = \tau^4 \quad (\text{penalty weight, } \rho = \sigma^2/q) \\
M &= \text{diag}(m), \quad D = D_2 = \text{the } (T-2) \times T \text{ second-difference matrix} \\
\mathcal{N}_m &= M + \rho D^T D \quad (\text{normal matrix of the masked fit}) \\
\hat{f} &= \mathcal{N}_m^{-1} M d \quad (\text{maximum-likelihood baseline, defined at every } t, \text{ masked or not}) \\
r &= M(d - \hat{f}) \quad (\text{the detrender's residual; zero at masked } t) \\
\lambda[t] &= (\mathcal{N}_m^{-1})_{tt} = \partial \hat{f}[t] / \partial d[t] \quad (\text{leverage, } \textit{detrending.pdf}, \text{ “Outputs”}) \\
R_m &= M - M \mathcal{N}_m^{-1} M \quad (\text{residual operator: } r = R_m d, \text{ and } \text{Cov}(r) = R_m^2 \text{ since } R_m \text{ is symmetric}) \\
S_0 &= (I + \rho D^T D)^{-1} \quad (\text{the unmasked smoother, } \mathcal{N}_m^{-1} \text{ at } m \equiv 1) \\
h[s] &= (S_0)_{t, t+s} \quad (\text{its impulse response, far from the ends}).
\end{aligned} \tag{1.1}$$

The unmasked smoother is a low-pass filter of unit mass, $\sum_s h[s] = 1$, and two of its numbers appear in every result below:

$$h[0] = \frac{c_2}{\tau}, \quad \sum_s h[s]^2 = \frac{c_h}{\tau}, \quad c_2 = \frac{1}{2\sqrt{2}} = 0.354, \quad c_h = \frac{3}{8\sqrt{2}} = 0.265. \tag{1.2}$$

c_2 is the flux-loss constant c_k of *detrending.pdf*, “The model” at $k = 2$; both constants are derived in §A.1, where the discrete values at $\tau = 64$ are shown to agree with them to 3×10^{-5} .

Peak-finding kernels. The reader is assumed to know *dedispersion.pdf*, “Peak-finding”. For $W_{\max} = 32$ there are $P = 3 \log_2 W_{\max} + 1 = 16$ profiles: $p = 0$ is the single sample [1], and each level $j = 0, \dots, 4$ contributes three, the boxcar of width 2^{j+1} and the two tapered kernels $[\frac{1}{2} \times 2^j, 1 \times 2^j, \frac{1}{2} \times 2^j]$ and $[\frac{1}{2} \times 2^j, 1 \times 2^{j+1}, \frac{1}{2} \times 2^j]$ (widths $3 \cdot 2^j$ and $4 \cdot 2^j$), with $p = 3j + q$ for $q = 1, 2, 3$. (The level is called λ in *dedispersion.pdf*, “Peak-finding”; here λ is the leverage, so the level is j .) The kernels are non-negative and unnormalized. We write

$$\begin{aligned} k_p &= \text{the kernel, as a vector; } k_{p,t} \text{ is the same vector placed at sample } t \\ W_p &= \text{its support width (1 to 64)} \\ W_{\text{eff},p} &= (\sum_i k_p[i])^2 / \|k_p\|^2 \quad (= W \text{ for a boxcar, } \frac{8}{3} 2^j \text{ and } \frac{18}{5} 2^j \text{ for the tapered kernels}) \\ y_p[t] &= \sum_i k_p[i] x[t-i] \quad (\text{the convolution of a residual } x \text{ with } k_p). \end{aligned} \tag{1.3}$$

Only variances are asked for, so the alignment convention of the convolution is immaterial. The three kernels quoted most often below are $p = 0$ (“p0”), the width-32 boxcar $p = 13$ (“box32”, $W_{\text{eff}} = 32$) and the widest tapered kernel $p = 15$ (“w64”, $W = 64$, $W_{\text{eff}} = 57.6$). The kernels are *not* narrow compared with the smoother: W_{eff}/τ runs from $1/64$ (p0) through $1/8$ (box8) to 0.5 (box32) and 0.9 (w64). Several results below have a simple form to first order in W_{eff}/τ ; those leading-order formulas are quoted for orientation only, and an exact number is given for every kernel. At this τ the unmasked detrender by itself already removes a large part of the wide kernels’ variance: $U_p/\|k_p\|^2 = 0.78$ (box32) and 0.63 (w64) (Table 1).

The quantities. For a fixed mask, averaging over the noise (and over any simulated noise the algorithm draws),

$$\begin{aligned} U_p &= \text{Var}(y_p[t]) \text{ with no mask and } x = r \quad (\text{independent of } t; \text{ “unmasked”}) \\ V_p(t) &= \text{Var}(y_p[t]) \text{ with the mask and } x = r \quad (\text{the detrender as it stands}) \\ V'_p(t) &= \text{Var}(y_p[t]) \text{ with the mask and } x = r' \quad (r' = \text{the inpainted residual defined next}). \end{aligned} \tag{1.4}$$

The question is how far $V'_p(t)/U_p$ can be from 1 when the mask and (p, t) are chosen adversarially. Since $k_{p,t}$ is a vector, everything is a quadratic form; e.g. $U_p = \|(I - S_0)k_{p,t}\|^2$.

Table 1 gives U_p for all 16 kernels, both as defined and normalized by its value at $\tau = \infty$. At $\tau = \infty$ the smoother subtracts only a global straight line (the null space of D), which has no effect on a compact kernel far from the ends, so $U_p(\tau=\infty) = \|k_p\|^2$, and the normalized value $U_p/U_p(\tau=\infty)$ is the fraction of the kernel’s variance that the unmasked detrender leaves: the fractional impact of detrending on U_p , before any mask is applied.

1.2 Algorithms

Algorithm 0 (plain inpainting).

1. Run the masked detrender once: $\hat{f} = \mathcal{N}_m^{-1} M d$ at every t , and $r = M(d - \hat{f})$.
2. Draw $n'[t] \sim \mathcal{N}(0, \sigma^2)$ i.i.d., independent of d , and form $d' = M d + (I - M)(\hat{f} + n')$: valid samples keep their data, masked samples get the baseline plus simulated noise.
3. Run the detrender on d' with *no* mask: $r' = (I - S_0) d'$.
4. Convolve with the peak-finding kernels, $y'_p = k_p * r'$.

Algorithm A (leverage-enlarged inpainting).

1. Run the masked detrender once with the original mask: $\hat{f} = \mathcal{N}_m^{-1} M d$ at every t , $r = M(d - \hat{f})$, and the leverage $\lambda[t] = (\mathcal{N}_m^{-1})_{tt}$ at every valid t .

p	name	W_p	$U_p(\tau=\infty) = \ k_p\ ^2$	U_p	$U_p/U_p(\tau=\infty)$
0	p0	1	1	0.993	0.993
1	box2	2	2	1.972	0.986
2	w3	3	1.5	1.472	0.982
3	w4	4	2.5	2.438	0.975
4	box4	4	4	3.890	0.972
5	w6	6	3	2.890	0.963
6	w8	8	5	4.752	0.950
7	box8	8	8	7.559	0.945
8	w12	12	6	5.559	0.927
9	w16	16	10	9.011	0.901
10	box16	16	16	14.24	0.890
11	w24	24	12	10.25	0.854
12	w32	32	20	16.10	0.805
13	box32	32	32	25.10	0.784
14	w48	48	24	17.22	0.717
15	w64	64	40	25.13	0.628

Table 1: The unmasked variance $U_p = \|(I - S_0)k_p\|^2$ of (1.4) for the 16 peak-finding kernels at $k = 2$, $\tau = 64$, $W_{\max} = 32$ (“wn” is the tapered kernel of width n , “box n ” the boxcar; W_p is the support width). At $\tau = \infty$ the detrender has no effect on a compact kernel, $U_p(\tau=\infty) = \|k_p\|^2$, so the last column is the fraction of the kernel’s variance that the unmasked detrender leaves. Note that w64 and box32 have nearly the same U_p at this τ .

2. Enlarge the mask by dropping valid samples whose leverage is too high:

$$m_2[t] = m[t] \wedge (\lambda[t] \leq \lambda_{\text{thr}}), \quad \lambda_{\text{thr}} = \mu \frac{c_2}{\tau}, \quad M_2 = \text{diag}(m_2) \leq M. \quad (1.5)$$

The threshold is a multiple μ of the interior leverage $h[0] = c_2/\tau$: $\mu = 1$ is the leverage in an unmasked stretch, $\mu = 4$ the leverage at the edge of a semi-infinite gap (to 1%: 3.96 at $\tau = 64$), and $\mu = \infty$ drops nothing.

3. Draw $n'[t] \sim \mathcal{N}(0, \sigma^2)$ i.i.d., independent of d , and form $d' = M_2 d + (I - M_2)(\hat{f} + n')$: samples valid in m_2 keep their data; samples masked in m_2 , whether originally masked or dropped in step 2, get the *first-pass* baseline plus simulated noise.
4. Run the detrender on d' with *no* mask: $r' = (I - S_0) d'$.
5. Convolve with the peak-finding kernels, $y'_p = k_p * r'$.

With $\mu = \infty$ this is algorithm 0.

Algorithm C (A with the dropped residuals removed).

1–3. As in algorithm A.

4. Let $u = (M - M_2)r$ be the first-pass residual on the samples dropped in step 2 (zero elsewhere). Run the detrender on d' with no mask, and subtract the smoothed u :

$$r'_C = (I - S_0) d' - S_0 u. \quad (1.6)$$

This needs only one unmasked detrender run, not two.¹

¹Since $(I - S_0)u - u = -S_0 u$, (1.6) can be rewritten as $r'_C = (I - S_0)(d' + u) - u$: run the unmasked detrender once on $d' + u$, then subtract u from its residual, which touches only the dropped samples. So algorithm C costs the same as algorithm A, and the detrender need not process two data vectors. The input $d' + u$ has a simple meaning too: on a dropped sample $\hat{f} + n' + r = d + n'$, so $d' + u = M d + (I - M)(\hat{f} + n') + (M - M_2)n'$ is algorithm 0’s inpainted data with fresh simulated noise added on the dropped samples; subtracting u afterwards removes their first-pass residual from the output. Equation (1.6) remains the definition because it makes the identity (1.7) for C immediate.

5. Convolve with the peak-finding kernels, $y'_p = k_p * r'_C$.

(A fourth variant, algorithm B, re-runs the first pass with the enlarged mask and inpaints with the new baseline. It is strictly worse than A and C; see §A.9.)

1.3 Summary of results

Numbers are for $\tau = 64$; each is labelled as a theorem (proved below) or as numerical (exact linear algebra for a given mask, with adversarial masks found by simulated annealing, so “worst found” rather than “worst possible”). Expressions to first order in W_{eff}/τ are quoted for orientation only (§1.1); every quoted number is exact for the kernel named.

1. **Exact identities** (theorem, §A.2). With r the first-pass residual,

$$\begin{aligned} \text{algorithm 0: } r' &= r + (I - S_0)(I - M) n', \\ \text{algorithm A: } r' &= M_2 r + S_0(M - M_2) r + (I - S_0)(I - M_2) n', \\ \text{algorithm C: } r'_C &= M_2 r + (I - S_0)(I - M_2) n'. \end{aligned} \tag{1.7}$$

The second pass returns the first-pass residual on every sample it kept, plus high-passed simulated noise on the samples it inpainted. Inpainting therefore restores the masked samples’ own variance and *cannot repair a variance deficit on valid samples*: whatever the masked first pass removed from them through its leverage stays removed. The variances follow exactly, e.g. for algorithm 0

$$V'_p(t) = \|R_m k\|^2 + \|(I - M)(k - S_0 k)\|^2, \quad V_p(t) = \|R_m k\|^2, \quad U_p = \|(I - S_0)k\|^2 \quad (k = k_{p,t}). \tag{1.8}$$

2. **Upper side** (theorem, §A.3). For algorithms 0 and C and every mask, $V'_p/U_p \leq B_{1,p}/U_p$ with $B_{1,p} = \sum_s \max(k_s^2, (k_s - (S_0 k)_s)^2)$: 1.404 for box32, 1.804 for w64, 1.011 for p0; for A the bound adds $\|S_0 k\|^2/U_p$ (0.17, 0.37, 0.004). The leading-order form $1 + 2c_2 W_{\text{eff}}/\tau$ (1.354, 1.636) no longer even bounds the exact one. The bound is not approached: the largest value found by annealing (algorithm 0) is $1 + 0.050 W_{\text{eff}}/\tau = 1.025$ (box32), $1 + 0.056 W_{\text{eff}}/\tau = 1.051$ (w64) and 1.0003 (p0), from a sparse comb around the kernel plus a masked annulus. The upper side of plain inpainting is thus a few percent, an order of magnitude below the bound. With enlargement it is larger: algorithm C exceeds 1 by 12% (box32) and 19% (w64) for a kernel on a kept run beside enlarged samples (§A.6; confirmed by Monte Carlo), and a dedicated maximization (§A.11) finds larger values at small μ : $V'/U - 1$ up to 0.083 / 0.18 at $\mu = 2$ and 0.136 / 0.28 at $\mu = 1.2$ for box32 / w64, still inside the bound.

3. **Lower side of algorithm 0** (numerical, §A.4). With at most B masked samples near the kernel (“window” = the whole $\pm 24\tau$ search window, 3073 samples), the worst found is

B	32	64	128	256	512	1024	2048	window	unbounded
box32	0.871	0.871	0.802	0.515	0.225	0.140	0.130	0.130	0.129
w64	0.946	0.788	0.774	0.556	0.390	0.306	0.296	0.296	0.295
p0	0.9959	0.9959	0.991	0.964	0.844	0.451	0.066	0.015	0.0042

Up to $B \approx 1.5\tau$ (2τ for w64) the worst case is the kernel exactly filling a gap of its own width, $V'/U = 0.871$ (box32), 0.788 (w64), 0.996 (p0) (leading order $1 - c_h W_{\text{eff}}/\tau$: 0.867, 0.761, 0.996); beyond that it is an *island*, the kernel’s support left valid with gaps of $B/2$ on each side, whose mean the first pass absorbs. Half the leverage ($V'/U = 0.56$) is reached at two-sided gaps of 117 samples around the 32-sample support, and a kernel simply touching a free gap is already there ($V'/U = 0.60$ for box32, 0.57 for w64, i.e. a 40–43% loss; 0.86 for box8, 0.98 for p0). The floors are 0.129 (box32; $c_h W/\tau = 0.133$), 0.295 (w64: a new structure, the 32-sample unit-weight core left valid with the

shoulders masked; 0.371 if the whole support is left valid) and 0.0042 (p0). Random masks that leave every valid run long compared with τ are harmless, where “long” now means $\gtrsim 4\tau$.

4. **Lower side with enlargement.** For algorithm C the leverage threshold gives a rigorous bound (theorem, §A.5), for every mask,

$$\frac{V_p'(t)}{U_p} \geq \text{LB}_p(\mu) = 1 - [2(\mu - 1)c_2 + c_h] \frac{W_{\text{eff},p}}{\tau} + O\left(\frac{W^2}{\tau^2}\right), \quad (1.9)$$

where $\text{LB}_p(\mu)$ is the exact expression (A.9) and the expansion is for orientation only (it is off by up to 0.12 and goes negative for box32 at $\mu \geq 3.5$, w64 at $\mu \geq 2.4$). Exactly, at $\mu = 1.2, 1.5, 2, 3, 4$, LB is 0.791, 0.688, 0.533, 0.281, 0.109 for box32 and 0.608, 0.435, 0.211, 0.003, 0 for w64, and the bound vanishes identically once $\mu c_2 W_{\text{eff}}/\tau \geq 1$ (w64 at $\mu \geq 3.14$, box32 at $\mu \geq 5.66$). It is weak: the worst masks found are 0.08–0.42 above it. For algorithm A there is no bound of this form: the term $S_0(M - M_2)r$ in (1.7) costs about $0.5 W_{\text{eff}}/\tau$ (0.25 for box32, 0.35 for w64) for a kernel on a kept run between enlarged zones, independently of μ up to $\mu \approx 2$ (§A.6). A nevertheless stays above $\text{LB}(\mu)$ in every run (by ≥ 0.05): the b -term is present at full strength, but on those masks the a -part $\|R_m M_2 k\|^2$ is far from the extremal value the bound assumes. Its worst cases are numerical only. Table 2 gives the worst found for both, with the bound and the cost of the enlargement.

5. **Cost of the enlargement** (numerical, §A.7). On an isolated gap the rule masks extra samples on each side, out to where the leverage falls below threshold: for a 2048-sample gap (32τ), 72/50/31/10/0 samples per side at $\mu = 1.2/1.5/2/3/4$ (a free end: 72/51/33/13/0), nothing for gaps up to 128 once $\mu \geq 2$, up to 512 once $\mu \geq 3$, and nothing at all for $\mu \geq 4$. A region of valid density n has leverage $\approx n^{-3/4} c_2/\tau$ (to 0.3% in the mean), so everything below density $\mu^{-4/3}$ is masked wholesale (0.78/0.58/0.40/0.23/0.16). On a random mask of gaps with lengths log-uniform in $[1, 512]$ covering 10% of the samples, 6.3/2.8/0.9/0.02/0.01% of the valid samples are lost (1.5/0.8/0.4/0.08/0% for lengths in $[1, 4096]$).
6. **Verification** (§A.8). Every formula was checked against a Monte Carlo of the literal procedure at $\tau = 64$: for algorithm 0, 143632 (kernel, position) pairs over eleven masks, with normally distributed pulls; for A and C, 128 (kernel, variant) pairs on the four worst masks at $\mu = 2$.

	algorithm A, worst found			algorithm C, worst found			C: bound $\text{LB}(\mu)$			cost		
μ	box32	w64	p0	box32	w64	p0	box32	w64	p0	edge	$n <$	RFI
1.2	0.842	0.793	0.9940	0.871	0.789	0.9959	0.791	0.608	0.9936	72	0.78	6.3%
1.5	0.789	0.723	0.9926	0.871	0.788	0.9959	0.688	0.435	0.9903	50	0.58	2.8%
2	0.701	0.594	0.9894	0.781	0.629	0.9937	0.533	0.211	0.9848	31	0.40	0.9%
3	0.514	0.386	0.9828	0.522	0.388	0.9851	0.281	0.003	0.9739	10	0.23	0.02%
4	0.313	0.360	0.9755	0.309	0.360	0.9756	0.109	0	0.9629	0	0.16	0.01%
∞	0.130	0.296	0.0148	0.130	0.296	0.0148	–	–	–	0	–	0

Table 2: Worst-case $V_p'(t)/U_p$ at $\tau = 64$ as a function of the enlargement threshold μ (eq. 1.5), found by simulated annealing over masks within $\pm 24\tau$ of the kernel (§A.8); the rigorous bound (A.9) in its exact form for algorithm C; and the cost of the enlargement on benign masks: extra samples masked on each side of a 2048-sample (32τ) gap (a free end expands by 72/51/33/13/0 instead), the valid density below which a random mask is masked wholesale, and the fraction of valid samples lost on a random RFI-like mask with gap lengths log-uniform in $[1, 512]$ covering 10% of the timestream (§A.7). $\mu = \infty$ is algorithm 0, at its window-limited value (the unbounded floors are 0.129/0.295/0.0042).

Bottom line. At $\tau = 64$ with $W_{\text{max}} = 32$, plain inpainting no longer even fixes the problem it was designed for: a kernel filling a gap of its own width comes back to 0.87 (box32) or 0.79 (w64), a 13–21% deficit, and a kernel beside a long gap is at 0.60. An adversary needs only a few hundred masked samples

($B = 256$, gaps of 2τ) to halve the variance of box32, and about 2000 to drive it to 13%. A per-sample leverage threshold lifts the floor, but not to a useful level for the wide kernels. With $\mu = 3$ –4 it costs nothing on benign masks and raises the floor from 0.13/0.30 to 0.31–0.52 (box32) / 0.36–0.39 (w64); with $\mu = 2$ it costs 31 samples per long-gap edge and all data below density 0.4, and gives 0.70 (A) or 0.78 (C) for box32 and 0.59/0.63 for w64; below $\mu \approx 1.5$ nothing more is gained (C is at the gap-filling floor 0.871/0.788, A saturates at 0.84/0.79 because of its $S_0(M - M_2)r$ term) while the cost doubles. No threshold reaches 0.90 for box32 or w64. Across the 16 kernels (§A.5), a guarantee of 0.95 from algorithm C is available only for $W \leq 8$ (box8 at $\mu \leq 2$; w12 only at $\mu \leq 1.2$), and 0.90 up to $W = 16$ at $\mu \leq 2$ (box16/w16 at $\mu \leq 1.9/2.1$; w24, w32 and w48 only at $\mu \leq 1.7, 1.6$ and 1.4); for A subtract the b -term, so 0.95 needs $W \leq 8$ at $\mu \lesssim 1.5$ and 0.90 holds for $W \leq 12$ at $\mu \leq 2$. The deficits are no longer small multiples of W_{eff}/τ : they are order one, and even in those units the coefficients of the $\tau = 512$ baseline survive only for box32 (to 10–20%), not for w64 (§A.10). The route to a real guarantee is the exact per- (p, t) variance, used to renormalize rather than to mask (§A.9); and the ratio W_{max}/τ itself deserves scrutiny, since the unmasked detrender already removes 22–37% of the wide kernels’ variance ($U_p/\|k_p\|^2 = 0.78/0.63$) before any mask is applied. These are pointwise statements; under the chunk-averaged figure of merit of §1.4 the lower side shrinks by an order of magnitude and the ranking of A and C reverses.

1.4 A weaker figure of merit?

Given an inpainting algorithm, consider two figures of merit:

$$\begin{aligned} X_{\text{hi}}(p) &\equiv \max_{m,t} \text{ReLU} \left(\frac{V'_p(t)}{U_p} - 1 \right) \\ X_{\text{lo}}(p) &\equiv \max_{m,t} \text{ReLU} \left(1 - \frac{V'_p(t)}{U_p} \right) \end{aligned} \quad (1.10)$$

where $\text{ReLU}(x) \equiv \max(x, 0)$ as usual. Intuitively, X_{hi} tells us how much V'_p can “overshoot” U_p , and X_{lo} tells us how much V'_p can “undershoot” U_p , if the mask m and time t are chosen adversarially.

Previous work has explored the values of $(X_{\text{hi}}, X_{\text{lo}})$ for inpainting algorithms 0, A, B, C, and for varying choices of the μ -parameter. So far, we’re finding that it’s (comparatively) easier to make X_{hi} small, and harder to make X_{lo} small.

Next, I’d like to explore the following question: suppose we leave the definition of X_{hi} unchanged, but weaken the definition of X_{lo} as follows:

$$\tilde{X}_{\text{lo}}(p) \equiv \max_{m,t} \text{ReLU} \left(1 - \frac{1}{T_c U_p} \sum_{t'=t}^{t+T_c-1} V'_p(t') \right) \quad (1.11)$$

That is, instead of evaluating V'_p at an adversarially chosen time t , we average it over an adversarially chosen time window $[t, t + T_c)$, where the “chunk size” $T_c = 1024$.

Please analogously explore values of $(X_{\text{hi}}, \tilde{X}_{\text{lo}})$ for inpainting algorithms 0, A, C, and for varying choices of the μ -parameter.

Results. Table 3 gives the answer for the two widest kernels; every entry is the best found by simulated annealing (exact for the mask exhibited, a lower bound on the true maximum), with $\tilde{X}_{\text{lo}}$ seeded by an exact scan over periodic masks. All narrow kernels are unproblematic under either figure: for $W \leq 8$ every entry of the corresponding table is below 0.03 for every algorithm and μ (box8: $X_{\text{hi}} \leq 0.006$ for algorithms 0 and A, ≤ 0.033 for C; $\tilde{X}_{\text{lo}} \leq 0.029$), and p0 is below 0.005 throughout. The six-kernel version is Table 7, and the evaluator, the searches, the worst-mask structures, the Monte Carlo checks and the caveats are in §A.11.

alg.	μ	box32			w64		
		X_{hi}	X_{lo}	$\tilde{X}_{lo}$	X_{hi}	X_{lo}	$\tilde{X}_{lo}$
0	∞	0.025	0.87	0.086	0.050	0.70	0.123
A	1.2	0.021	0.16	0.035	0.042	0.21	0.047
A	1.5	0.022	0.21	0.047	0.045	0.28	0.068
A	2	0.023	0.30	0.069	0.048	0.41	0.100
A	3	0.024	0.49	0.077	0.050	0.61	0.118
A	4	0.025	0.69	0.083	0.050	0.64	0.123
C	1.2	0.136	0.13	0.013	0.280	0.21	0.009
C	1.5	0.111	0.13	0.044	0.218	0.21	0.060
C	2	0.083	0.22	0.068	0.180	0.37	0.098
C	3	0.057	0.48	0.077	0.109	0.61	0.117
C	4	0.043	0.69	0.083	0.087	0.64	0.123

Table 3: the figures of merit at $\tau = 64$, $T_c = 1024$, for algorithm 0 and for A and C at five thresholds μ (eq. 1.5). X_{lo} is the pointwise figure (1– the worst V'/U of Table 2), shown for comparison with its windowed replacement $\tilde{X}_{lo}$.

What the table says:

1. **Averaging over a chunk does most of the work.** For algorithm 0, $\tilde{X}_{lo}$ is 0.086 (box32) and 0.12 (w64) against $X_{lo} = 0.87$ and 0.70. A masked stretch comes back at $V' = U$ exactly (§A.2), so the thousands of masked samples the pointwise adversary uses to isolate a kernel count as 1 in the average; the windowed adversary is left with “sparse islands” (valid runs of 1–1.5 W separated by gaps of 2–3 τ , valid fraction ~ 0.2), trading the depth of the deficit against the fraction of the window it occupies.
2. **The leverage rule helps the windowed figure only modestly.** It masks everything below valid density $\mu^{-4/3}$ wholesale, so the windowed adversary keeps its density just above that, where nothing is enlarged and A, C and algorithm 0 coincide. At $\mu = 2$ the gain is 20% (0.086 $\rightarrow$ 0.069 for box32); at $\mu \geq 3$ it is nil; at $\mu = 1.2$ it is larger (0.035 for A, 0.013 for C), at the overmasking cost of Table 2.
3. **On the upper side A is as good as algorithm 0 and C is not.** X_{hi} of A is 0.02–0.025 (box32) and 0.04–0.05 (w64) for every μ , because its extra term is a negative cross-covariance. A dedicated maximization for C finds 0.08 / 0.18 at $\mu = 2$ and 0.14 / 0.28 at $\mu = 1.2$ (a short kept run inside the support with everything around it enlarged; §A.11), far above the 12–19% noted incidentally in §1.3.
4. **Verdict.** Under $(X_{hi}, \tilde{X}_{lo})$, A dominates C: equal or within 0.01 on the lower side for $\mu \geq 1.5$, and 3–6 $\times$ smaller on the upper side. A at $\mu = 1.5$ –2 gives (0.022, 0.05–0.07) for box32 and (0.045, 0.07–0.10) for w64, i.e. both sides at the few-percent level, for 31–50 extra masked samples per long-gap edge and 1–3% of the valid samples on the RFI-like mask of Table 2. C’s small $\tilde{X}_{lo}$ at $\mu = 1.2$ is bought with a 14–28% overshoot.
5. **Caveats.** The values are lower bounds on the true maxima (annealing). T_c matters little: for the periodic family the algorithm-0 maximum is 0.101, 0.076, 0.069, 0.068 (box32) at $T_c = 256, 1024, 4096, \infty$, so $T_c = 1024$ is within 10% of the large- T_c limit and the reduction is the averaging itself, not the chunk size.

A 1-d Kalman inpainting: proofs and details

This appendix contains the proofs and the numerical details behind §1, in the notation of §1.1.

A.1 More notation, and the constants of the smoother

The hat matrix and the correlation length. Two more objects from *detrending.pdf*, “Time detrending algorithm 2: Kalman filter” are used in the proofs:

$$\begin{aligned} L_m &= M \mathcal{N}_m^{-1} M \quad (\text{masked hat matrix, so that } R_m = M - L_m) \\ \ell &= \sqrt{2} \tau = 90.5 \quad (\text{correlation length: entries of } \mathcal{N}_m^{-1} \text{ decay as } e^{-|t-t'|/\ell}). \end{aligned} \quad (\text{A.1})$$

L_m is the hat matrix $H = \mathcal{N}_m^{-1} M$ of *detrending.pdf*, “Outputs” restricted to the valid samples, made symmetric, and $\lambda[t] = (L_m)_{tt}$ at valid t . $\mathcal{N}_m$ is positive definite as soon as two samples are valid. The $r_{\min}$ mask expansion of *detrending.pdf*, “Outputs” is ignored throughout; it fires only at the start of a stream.

The unmasked smoother. S_0 reproduces cubics: its normal equations $(I + \rho D^T D)f = d$ hold with equality for any f with vanishing fourth difference. In particular $\sum_s h[s] = 1$. Far from the ends, S_0 is Toeplitz with spectrum $s(\omega) = 1/(1 + \rho(2 - 2\cos\omega)^2)$, which for $\omega\tau \ll 1$ is $1/(1 + (\omega\tau)^4)$. Then $h[0] = \int \frac{d\omega}{2\pi} s = \frac{1}{\tau} \frac{1}{2\pi} \frac{\pi}{\sqrt{2}} = \frac{c_2}{\tau}$ and $\sum_s h[s]^2 = \int \frac{d\omega}{2\pi} s^2 = \frac{1}{\tau} \frac{1}{2\pi} \frac{3\pi}{4\sqrt{2}} = \frac{c_h}{\tau}$, which is (1.2); the discrete values at $\tau = 64$ agree to 3×10^{-5} ($h[0]$) and 1×10^{-5} ($\sum h^2$), and at $\tau = 512$ to 5×10^{-7} . In units of ℓ the shape of h is scale-invariant: it is positive and decreasing out to $|s| = 2.36\ell$ (3.3τ), with $h(\ell)/h[0] = 0.51$ and $h(2\ell)/h[0] = 0.067$, beyond which comes a first negative lobe of depth 4.3% of $h[0]$ (at 3.1ℓ) and integrated mass -0.07 of the total. For a kernel narrow compared with ℓ the smoothed kernel is flat across the support, $(S_0 k)_s \approx h[0] \sum k$; at $\tau = 64$ it is flat to 2.4% (box32) and 8.7% (w64), and $h[0] \sum k$ overestimates its maximum by 1% and 2.7%. To leading order in W/τ ,

$$\langle k, S_0 k \rangle \approx \frac{c_2 (\sum k)^2}{\tau}, \quad \|S_0 k\|^2 \approx \frac{c_h (\sum k)^2}{\tau}, \quad U_p = \|k\|^2 - 2\langle k, S_0 k \rangle + \|S_0 k\|^2 \approx \|k\|^2 \left(1 - 0.442 \frac{W_{\text{eff},p}}{\tau}\right). \quad (\text{A.2})$$

This is why $W_{\text{eff}} = (\sum k)^2 / \|k\|^2$, and not the support width, is the parameter that appears everywhere. The leading-order U_p is accurate to 0.1% up to box16 and is 0.7% (box32), 1.7% (w48) and 4.3% (w64) low (the $O(W^2/\tau^2)$ term is positive); the exact U_p of every kernel, and its ratio to $\|k\|^2$, are in Table 1. The kernel constants that enter (A.2) are:

	p0	box2	w3	w4	box4	w6	w8	box8
$\ k\ ^2$	1	2	1.5	2.5	4	3	5	8
W_{eff}	1	2	2.67	3.6	4	5.33	7.2	8
	w12	w16	box16	w24	w32	box32	w48	w64
$\ k\ ^2$	6	10	16	12	20	32	24	40
W_{eff}	10.7	14.4	16	21.3	28.8	32	42.7	57.6

(“wn” is the tapered kernel of width n , “box n ” the boxcar.)

A.2 The exact identities

Algorithm 0. Since $Md = M\hat{f} + r$, the inpainted data are $d' = \hat{f} + r + (I - M)n'$. The normal equations $\mathcal{N}_m \hat{f} = Md$ say $\rho D^T D \hat{f} = Md - M\hat{f} = r$. And the unmasked smoother satisfies $I - S_0 = S_0(S_0^{-1} - I) = \rho S_0 D^T D$. Hence

$$(I - S_0)\hat{f} = \rho S_0 D^T D \hat{f} = S_0 r, \quad (I - S_0)(\hat{f} + r) = S_0 r + r - S_0 r = r, \quad (\text{A.3})$$

and $r' = (I - S_0)d' = r + (I - S_0)(I - M)n'$. The proof uses exactly: the same ρ (same τ) and the same D (same k and the same diffuse treatment of the ends) in both passes, whole-timestream solves so that the normal equations hold, masked samples filled with the first-pass $\hat{f}$ at those samples (plus anything independent), and an unmasked second pass. Nothing is assumed about d or n' ; the identity is deterministic. Numerically it holds to 3×10^{-13} ; with ρ changed by a factor 2 in the second pass it fails at the 5% level.

Algorithm A. $M_2 \leq M$ gives $M_2 r = M_2(d - \hat{f})$, so $M_2 d = M_2 \hat{f} + M_2 r$ and $d' = \hat{f} + M_2 r + (I - M_2)n'$. Then

$$(I - S_0)(\hat{f} + M_2 r) = S_0 r + M_2 r - S_0 M_2 r = M_2 r + S_0(M - M_2)r \quad (\text{A.4})$$

using $r = Mr$. The middle term is a smoothed copy of the residuals at the samples the enlargement dropped: their data are gone from d' , but their imprint on $\hat{f}$ is not.

Algorithm C subtracts exactly that term, $S_0 u$ with $u = (M - M_2)r$, leaving $r'_C = M_2 r + (I - S_0)(I - M_2)n'$.

Algorithm B (first pass re-run with m_2 , baseline $\hat{f}_2 = \mathcal{N}_{m_2}^{-1} M_2 d$, residual $r_2 = M_2(d - \hat{f}_2)$) is algorithm 0 with $m \rightarrow m_2$, so $r' = r_2 + (I - S_0)(I - M_2)n'$.

Variances. $r = R_m d$ with $R_m = M - M \mathcal{N}_m^{-1} M$ symmetric, so $\text{Cov}(r) = R_m^2$; n' is independent of r ; and S_0 is symmetric. For a kernel vector $k = k_{p,t}$,

$$\begin{aligned} 0: & V_p'(t) = \|R_m k\|^2 + \|(I - M)(k - S_0 k)\|^2, \\ A: & V_p'(t) = \|R_m \tilde{k}\|^2 + \|(I - M_2)(k - S_0 k)\|^2, \quad \tilde{k} = M_2 k + (M - M_2)S_0 k, \\ C: & V_p'(t) = \|R_m M_2 k\|^2 + \|(I - M_2)(k - S_0 k)\|^2, \\ B: & V_p'(t) = \|R_{m_2} k\|^2 + \|(I - M_2)(k - S_0 k)\|^2, \end{aligned} \quad (\text{A.5})$$

with $U_p = \|(I - S_0)k\|^2$ and $V_p(t) = \|R_m k\|^2$. The first term in each line is the contribution of the first-pass residual; the second is the simulated noise, high-passed and restricted to the inpainted samples.

A.3 Upper bounds

Lemma. $0 \leq L_m \leq M$, hence $0 \leq R_m \leq M \leq I$ and $R_m^2 \leq R_m$. *Proof.* $L_m = M \mathcal{N}_m^{-1} M \geq 0$ is clear. For $w = Mw$, the variational form $w^T \mathcal{N}_m^{-1} w = \max_v (2v^T w - v^T \mathcal{N}_m v)$ and $\mathcal{N}_m \geq M$ give $w^T \mathcal{N}_m^{-1} w \leq \max_v (2v^T w - v^T M v) = \max_u (2u^T w - u^T u) = w^T w$ with $u = Mv$. So $v^T L_m v \leq v^T M v$ for every v . (At $v = e_t$ this is the $\lambda[t] \leq 1$ of *detrending.pdf*, “Outputs”.) $\square$

Algorithm 0. By the lemma $\|R_m k\|^2 = k^T R_m^2 k \leq k^T M k = \|M k\|^2$, so

$$\begin{aligned} V_p'(t) & \leq \|M k\|^2 + \|(I - M)(k - S_0 k)\|^2 = \sum_s \left[m_s k_s^2 + (1 - m_s)(k_s - (S_0 k)_s)^2 \right] \\ & \leq \sum_s \max(k_s^2, (k_s - (S_0 k)_s)^2) =: B_{1,p}. \end{aligned} \quad (\text{A.6})$$

On the support $0 \leq (S_0 k)_s \leq 0.258 \leq k_s < 2k_s$ (the maximum of $(S_0 k)_s$ over all s lies on the support for every kernel, and is 0.258 for w64 against $k_s \geq \frac{1}{2}$: the condition holds for all 16 kernels, with little room to spare), so the maximum is k_s^2 there; off the support it is $(S_0 k)_s^2$. Hence $B_{1,p} \leq \|k\|^2 + \|S_0 k\|^2$, and with (A.2)

$$\frac{V_p'(t)}{U_p} \leq \frac{B_{1,p}}{U_p} \leq 1 + \frac{2\langle k, S_0 k \rangle}{U_p} = 1 + 2c_2 \frac{W_{\text{eff}}}{\tau} + O\left(\frac{W^2}{\tau^2}\right). \quad (\text{A.7})$$

Exactly, $B_{1,p}/U_p = 1.404$ (box32), 1.804 (w64), 1.011 (p0); the loosened form $1 + 2\langle k, S_0 k \rangle/U_p$ is 1.443, 1.964, 1.011; and the leading-order expression $1 + 2c_2 W_{\text{eff}}/\tau$ is 1.354, 1.636, 1.011, *below* the exact bound by 4–17% (the $O(W^2/\tau^2)$ term is +0.05 for box32 and +0.17 for w64), so at $\tau = 64$ only the exact form is the content of the theorem. A tighter bound, $1 + c_2 W_{\text{eff}}/\tau$ at leading order, follows from $\|R_m k\|^2 \leq k^T R_m k = \|M k\|^2 - (M k)^T \mathcal{N}_m^{-1} (M k)$ and $\mathcal{N}_m^{-1} \geq S_0$ (more mask, larger inverse), *provided* the resulting mask-dependent expression $B_2 = \max_m [\|M k\|^2 - (M k)^T S_0 (M k) + \|(I - M)(k - S_0 k)\|^2]$ is attained by the all-valid support. At $\tau = 64$ that proviso is false for the ten kernels of width ≥ 12 : masking 1–24 support samples (the half-weight shoulders of the tapered kernels, edge samples of the boxcars) raises the expression by up to 10^{-2} in the ratio. The maximum found by local search over support masks is $B_2/U_p = 1.184$ (box32), 1.352 (w64), 1.0055 (p0), a numerical statement only. Neither bound is approachable: saturating

them needs zero leverage on the kernel and every sample outside the support masked, and the second makes the support an island, whose mean mode has leverage near 1.

Algorithms A and C. For C the first term is $\|R_m M_2 k\|^2 \leq \|M_2 k\|^2$ and the same argument gives (A.7) unchanged. For A, $\|R_m \tilde{k}\|^2 \leq \|\tilde{k}\|^2 = \|M_2 k\|^2 + \|(M - M_2)S_0 k\|^2 \leq \|M_2 k\|^2 + \|S_0 k\|^2$ (the two pieces of $\tilde{k}$ have disjoint supports), which adds $\|S_0 k\|^2/U_p$ to the bound: 0.17 (box32), 0.37 (w64), 0.004 (p0), i.e. $c_h W_{\text{eff}}/\tau$ at leading order.

What is actually reached. Annealing on algorithm 0 (§A.8; 7 seeds of 90000 steps per kernel, agreeing to 3×10^{-4}) finds at most $1 + 0.050 W_{\text{eff}}/\tau = 1.025$ (box32), $1 + 0.056 W_{\text{eff}}/\tau = 1.051$ (w64) and 1.0003 (p0): an irregular sparse comb of 1–6-sample valid runs (valid fraction 0.3–0.5) within $\sim 1.5\tau$ of the kernel, a fully masked annulus from $\sim 2\tau$ to $\sim 6\tau$ where the smoother’s autocorrelation $(h * h)/h$ exceeds 1, and a valid patch of $\sim 0.9\tau$ beyond it (for p0 the annulus alone). The comb alone gives 1.022 (box32) and 1.045 (w64) in a sweep over periods 2–128 and densities $1/8$ – $7/8$ (the maximum is at $n = 3/8$); the homogenized-smoother formula $1 + 0.442 n(1 - n^{1/4}) W_{\text{eff}}/\tau$ of §A.4 ($\leq 1 + 0.036 W_{\text{eff}}/\tau$ at $n = 0.41$, i.e. 1.018/1.032) is 20–40% low at $\tau = 64$, where it was good to 10^{-4} at $\tau = 512$; the annulus adds the rest (for box32 the near part alone gives 1.022, the annulus alone 1.006). With enlargement switched on, algorithm C reaches 1.12 (box32) and 1.19 (w64) on the structured run and edge masks of §A.6, inside the theorem; a dedicated maximization (§A.11) finds larger values at small μ , 1.083 / 1.18 at $\mu = 2$ and 1.136 / 1.28 at $\mu = 1.2$ for box32 / w64 (a short kept run inside the support with everything within 3 – 4τ enlarged), still inside the bound.

A.4 Plain inpainting: the lower side

By (1.8), $V'_p \geq \|(I - M)(k - S_0 k)\|^2$, and every deficit is in the first term $\|R_m k\|^2 = \|Mk\|^2 - 2k^T L_m k + \|L_m k\|^2$: the masked first pass shrinks the data toward its baseline, and the kernel loses whatever part of itself the baseline can follow. Writing $\lambda_k = k^T L_m k / \|Mk\|^2$ for the leverage on the kernel, $\|R_m k\|^2 / \|Mk\|^2 = 1 - 2\lambda_k + \nu_k$ with $\nu_k = \|L_m k\|^2 / \|Mk\|^2 \leq \lambda_k$, the kernel-level version of the residual variance of *detrending.pdf*, “Outputs”. With $\bar{\lambda}$ the per-sample leverage averaged under the kernel, $\lambda_k = W_{\text{eff}} \bar{\lambda}$ to 0.1% for $W \leq 8$ in every configuration examined; for box32 λ_k is 1–2% below $W_{\text{eff}} \bar{\lambda}$ and for w64 5–9% below, because L_m is no longer flat on the scale of the kernel (0.35 – 0.7ℓ wide). The per-sample leverage of *detrending.pdf*, “Outputs” is still the first-order quantity; what separates benign from harmful configurations at equal leverage is the recovery term ν_k , which is 0.68–0.90 of λ_k in the configurations examined (0.76 in the unmasked interior for box32), far from the $L_m k = \lambda k$ idealization ($\nu = \lambda^2$). The rest of this subsection is numerical (exact linear algebra at $\tau = 64$ unless stated).

Islands. Let the kernel’s support be valid, with gaps $G_1 \leq G_2$ on either side and valid data beyond. For $G \gg W$ the baseline inside a gap is a cubic (the smoother reproduces cubics), the penalty $\rho \sum (\Delta^2 f)^2$ is a bending energy, and displacing the island by δ costs $c \rho \delta^2 / G_1^3$ with $c = 3(1 + G_1/G_2)^3$ (24 for equal gaps, the clamped beam with a central load; 3 for $G_2 \rightarrow \infty$, the cantilever). An island of A valid samples then has its mean mode fitted with leverage

$$\lambda_A = \frac{A}{A + K}, \quad K = 3\rho \left(\frac{1}{G_1 + \ell} + \frac{1}{G_2 + \ell} \right)^3, \quad (\text{A.8})$$

where the $+\ell$ is an empirical correction (the valid data clamp the baseline about ℓ inside their edge, not at it) that makes the fitted c a universal function of G/τ : for $A = 1$ the fitted values at $\tau = 64$ (1.21, 4.20, 9.31, 14.6, 18.6, 21.1 at $G/\tau = 1, 2, \dots, 32$) are those at $\tau = 512$, and the law is exact. For $A = 32$ and 64 ($A/\tau = \frac{1}{2}, 1$) it under-estimates the leverage by 10–20% for $G \lesssim 8\tau$ (at $G = 2\tau$: 0.53 exact against 0.45 for $A = 32$, 0.73 against 0.62 for $A = 64$; at $G = 4\tau$, 0.80 against 0.77 and 0.90 against 0.87) and is exact for $G \geq 16\tau$; the excess is the finite width of the island relative to ℓ (the beam’s point load is wrong), not the slope mode, whose leverage stays below 0.05 for $A = 32$ but reaches 0.15–0.41 for $A = 64$ at $G \geq 16\tau$, where it matters for the w64 floor. For the boxcar matched to the island, $L_m k \approx \lambda_A k$ would give $V_p \approx W(1 - \lambda_A)^2$ and $V'_p/U_p \approx (1 - \lambda_A)^2(1 + 0.442 W/\tau) + c_h W/\tau$; at $\tau = 64$ this underestimates V'/U by 0.02 ($A = 1$) to 0.18 ($A = 64$) at $\lambda_A \approx \frac{1}{2}$, since $\nu_k = 0.34$ against $\lambda_k^2 = 0.25$ on the tuned 32-island. Half the leverage is reached

at $G_{1/2} = (c\rho/A)^{1/3} - \ell$ for $A = 1$ (647 two-sided, 277 one-sided; the formula gives 648 and 279) but not for $A = 32$: numerically $G_{1/2} = 117$ two-sided (formula 142), and one-sided about one sample (formula 26), because the 32-kernel simply touching a free gap already has $\lambda_A = 0.50$, so the one-sided island *is* the valid-side edge case below. Two-sided scan for box32 on its own support (V'/U): 0.930, 0.802, 0.560, 0.515, 0.225, 0.140, 0.130, 0.129 at $G = 32, 64, 117, 128, 256, 512, 1024, 2048$; for p0, 0.964, 0.844, 0.451, 0.279, 0.066, 0.007 at $G = 128, 256, 512, 647, 1024, 2048$. As $G \rightarrow \infty$ the floor is the simulated-noise term: 0.129 for box32 ($c_h W/\tau = 0.133$ is 2.4% high, the support part of $\|S_0 k\|^2$ no longer being negligible), 0.033 (box8), 0.066 (box16), 0.0042 for p0 (c_h/τ to 0.1%). For the tapered kernels the island absorbs only the constant part of the kernel and the shoulders survive: the floor is $(\|k\|^2 - (\sum k)^2/W)/U_p + (n'\text{-term}) = 0.159 + 0.212 = 0.371$ for w64 on its full support (0.317 for w48) — but a worse structure exists: leaving only the 32 unit-weight core samples valid, with the shoulders masked along with the gaps, gives 0.295 (found by annealing at $B \geq 512$; at $B \leq 256$ the full-support island is still worse for the kernel). A kernel narrower than its island: the rule that it loses only the fraction W/A of its variance fails at $\tau = 64$ for $A \geq 48$, since an island wider than $\sim \tau$ supports many low-leverage modes; box32 centred on an island of $A = 48, 64, 128$ samples (gaps 4096) has $V'/U = 0.54, 0.73, 0.97$ (W/A rule: 0.45, 0.59, 0.79), and exceeds 1 at $A = 8\tau$. The isolation length scales as $\tau^{4/3}A^{-1/3}$ rather than as τ .

Sparse periodic masks (valid runs A , gaps B , density $n = A/(A+B)$, period $\ll \tau$). For modes smooth on the period, $\mathcal{N}_m \approx nI + \rho D^T D$: the masked smoother is the unmasked one with $\tau_{\text{eff}} = \tau n^{-1/4}$, and the per-sample leverage is $c_2 n^{-3/4}/\tau$. A kernel on one run has, to leading order, $V'/U = 1 - (W/\tau)[0.442(n^{-3/4} - 1) - 0.265(1 - n)]$; at $\tau = 64$ this is off by 0.02–0.08 for $n \geq 1/5$ and meaningless below $n \approx 1/8$ (the exact values saturate at the island floor 0.129 while the formula goes negative). Exactly, for $W = A = 32$: 0.894, 0.756, 0.624, 0.505, 0.265, 0.137 at $n = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{8}, \frac{1}{16}$; the deficit reaches 10% already at $n = 0.51$ (every other 32-run masked) and 50% at $n = 0.20$ (gaps of 2τ). For w64 on 64-runs, 10% at $n = 0.65$ and 50% at $n = 0.29$, floor 0.371. A kernel spanning many periods sees the comb regime, $V'/U = 1 + 0.442n(1 - n^{1/4})W_{\text{eff}}/\tau \geq 1$ at leading order: the excess is real for every $W \geq 2$ (p0 is below 1, 0.9976 at $n = \frac{1}{2}$), but the formula is 20–40% low, e.g. the every-other-sample comb gives 1.021 (box32), 1.043 (w64), 1.0046 (box8) against 1.018, 1.032, 1.0044.

Gap edges. The leverage at the edge of a semi-infinite gap is $3.96 c_2/\tau$ at $\tau = 64$ ($4c_2/\tau$ with a 1% finite- τ correction) and the corresponding ν is $3.97 c_h/\tau$; the excess decays into the valid data over 2–4 τ (50% of its edge value at 0.34τ , 10% at τ , 1% at 3.2τ), and the profile in units of τ is the same to three digits as at $\tau = 512$. A 32-wide kernel at the edge covers leverage from 3.96 down to $2.0 c_2/\tau$, so its kernel-averaged edge leverage is $2.9 c_2/\tau$. A kernel on the valid side, touching a free gap, has $V'/U = 0.599$ (box32), 0.568 (w64; 0.556 at its worst position, 16 samples into the gap), 0.865 (box8), 0.982 (p0), and inpainting does not help since nothing under the kernel is masked; the leading-order $1 - 1.1 W_{\text{eff}}/\tau$ is valid only for $W_{\text{eff}}/\tau \lesssim 0.15$ (0.863 for box8) and would give 0.45/0.01 for box32/w64. The deficit recovers over $\sim \tau/2$ as the kernel moves away from the edge (0.60, 0.72, 0.82, 0.93, 1.001 at distance 0, 8, 16, 32, 64 for box32), with a –1% dip at 2τ from the annulus of §A.3. Edge leverage for finite gaps, in units of c_2/τ : 1.02, 1.10, 1.42, 1.81, 2.37, 2.94, 3.64 at $G = 4, 16, 64, 128, 256, 512, 2048$ (scale-invariant: the $\tau = 512$ values at $8\times$ the gap); box32 touching a gap of $G = 64/256/1024$ is at 0.943/0.780/0.656. A kernel exactly filling a gap of its own width has $V = 0$ but $V'/U = 1 - \sum_{\text{valid}} (S_0 k)_s^2 / U_p = 0.871$ (box32), 0.788 (w64), 0.967 (box8), 0.996 (p0) (leading order $1 - c_h W_{\text{eff}}/\tau$: 0.867, 0.761): the simulated noise restores everything except the smoothing leakage. A kernel at the edge of a long gap, inside it, is at 0.935 (box32) / 0.894 (w64), and one width further in at 0.967/0.982.

Adversarial search. Annealing over masks within $\pm 24\tau$ of the kernel with a bound B on the number of masked samples (§A.8) found nothing beyond these structures: for $B \lesssim 1.5\tau$ (2τ for w64) the optimum is the gap of width W (for w64 at $B = 32 < W$, a sub-support gap over its 32 unit-weight core samples, 0.946), beyond that the symmetric island with gaps $B/2$ (one-sided or asymmetric islands and sparse periodic masks are always worse for the adversary), and the optimum matches the best analytic candidate to 10^{-5} at every budget, except that for w64 at $B \geq 512$ the annealer found the core-only island, 0.02–0.08 below the full-support island. This is the table in §1.3, item 3.

A.5 Algorithm C: the lower bound

Theorem. Let $\lambda_{\text{thr}} = \mu c_2 / \tau$, $k \geq 0$ a kernel with support supp , and $\mathcal{A} = \text{supp} \cap m_2$ the kept part of the support. For every mask,

$$V'_p(t) \geq \min_{\mathcal{A} \subseteq \text{supp}} \left[\|k_{\mathcal{A}}\|^2 (1 - \lambda_{\text{thr}} W_{\text{eff}}(k_{\mathcal{A}}))_+^2 + \sum_{s \in \text{supp} \setminus \mathcal{A}} (k_s - h[0] \sum k)^2 \right] =: U_p \text{LB}_p(\mu), \quad (\text{A.9})$$

where $k_{\mathcal{A}}$ is k restricted to $\mathcal{A}$ and $W_{\text{eff}}(k_{\mathcal{A}}) = (\sum k_{\mathcal{A}})^2 / \|k_{\mathcal{A}}\|^2$. For every one of the 16 kernels and every $\mu \geq 1$ the minimizer is $\mathcal{A} = \text{supp}$ (by enumeration at $\tau = 64$), and then $\text{LB}_p(\mu) = \|k\|^2 (1 - \mu c_2 W_{\text{eff}} / \tau)_+^2 / U_p$, whose expansion is (1.9). Where $\mu c_2 W_{\text{eff}} / \tau \geq 1$ the bound is identically zero: w64 for $\mu \geq 3.14$, w48 for $\mu \geq 4.24$, box32 for $\mu \geq 5.66$.

Proof. By (A.5), $V'_C = \|R_m k_{\mathcal{A}}\|^2 + \sum_{s \notin m_2} (k_s - (S_0 k)_s)^2$, since $M_2 k = k_{\mathcal{A}}$. (i) For v supported on valid samples, with $\lambda_v = v^T L_m v / \|v\|^2 \in [0, 1]$ (lemma of §A.3) and Cauchy-Schwarz $(v^T L_m v)^2 \leq \|v\|^2 \|L_m v\|^2$,

$$\|R_m v\|^2 = \|v\|^2 - 2v^T L_m v + \|L_m v\|^2 \geq \|v\|^2 (1 - \lambda_v)^2, \quad (\text{A.10})$$

and $(1 - \lambda)^2$ decreases on $[0, 1]$. (ii) L_m is positive semidefinite, so $|(L_m)_{ss'}| \leq \sqrt{\lambda_s \lambda_{s'}}$; for $v \geq 0$ supported on kept samples, all of which have $\lambda_s \leq \lambda_{\text{thr}}$, $v^T L_m v \leq (\sum_s v_s \sqrt{\lambda_s})^2 \leq \lambda_{\text{thr}} (\sum_s v_s)^2$, i.e. $\lambda_v \leq \lambda_{\text{thr}} W_{\text{eff}}(v)$. With (i), $\|R_m k_{\mathcal{A}}\|^2 \geq \|k_{\mathcal{A}}\|^2 (1 - \lambda_{\text{thr}} W_{\text{eff}}(k_{\mathcal{A}}))_+^2$. (iii) For $s \in \text{supp} \setminus \mathcal{A}$, $(S_0 k)_s = \sum_i h[s - i] k_i \leq h[0] \sum k \leq 0.265 \leq \frac{1}{2} \leq k_s$ (the first inequality since h is positive over the support; the numbers are those of w64, and there is no slack for a wider kernel), and $x \mapsto (k_s - x)^2$ decreases on $[0, k_s]$, so $(k_s - (S_0 k)_s)^2 \geq (k_s - h[0] \sum k)^2$ (if the negative lobes of h make $(S_0 k)_s < 0$ the term only grows). Masked samples outside the support contribute ≥ 0 . Summing gives (A.9). (iv) For a boxcar and $a = |\mathcal{A}|$ the bracket is $g(a) = a(1 - \lambda_{\text{thr}} a)^2 + (W - a)(1 - W h[0])^2$, with $g'' = -4\lambda_{\text{thr}} + 6\lambda_{\text{thr}}^2 a < 0$ for $a < 2/(3\lambda_{\text{thr}}) = 2\tau/(3\mu c_2)$, which covers $[0, W]$ for every boxcar at $\mu \leq 3$ (at $\tau = 64$, $a < 40$ at $\mu = 3$); then the minimum is at an endpoint, and $g(W) - g(0) = W[(1 - \mu W h[0])^2 - (1 - W h[0])^2] \leq 0$ for $\mu \geq 1$. Beyond that (box32 at $\mu \geq 3.8$: $a \geq 31$ at $\mu = 4$, $a \geq 21$ at $\mu = 6$) concavity fails on part of the range, and direct enumeration over $a = 0, \dots, W$ gives the same conclusion. For the tapered kernels the bracket depends only on how many unit- and half-weight samples are kept, and enumeration gives the same conclusion for every kernel and every $\mu \geq 1$ (at $\mu = 1$ exactly, $g(0) = g(W)$ to 3×10^{-5}). $\square$

Remarks. The bound cannot use the recovery term $\|L_m k\|^2$ (per-sample leverage bounds the hat matrix from above, not below), which is why $\text{LB}(1) = 1 - c_h W_{\text{eff}} / \tau$ at leading order rather than 1; it lies just below inpainting's gap-filling floor (0.864 against 0.871 for box32, 0.740 against 0.788 for w64). The leading-order form of (1.9) is within 10^{-4} of the exact LB for $W \leq 8$, but for box32 it is 3.5% low at $\mu = 2$ and meaningless at $\mu \geq 3$, and for w64 40% low at $\mu = 2$. Exact values for all 16 kernels:

μ	p0	box2	w3	w4	box4	w6	w8	box8
1	0.9959	0.9917	0.9890	0.9851	0.9834	0.9779	0.9702	0.9669
1.2	0.9936	0.9873	0.9831	0.9771	0.9746	0.9661	0.9542	0.9491
1.5	0.9903	0.9807	0.9742	0.9652	0.9614	0.9485	0.9304	0.9227
2	0.9848	0.9697	0.9596	0.9455	0.9395	0.9195	0.8915	0.8795
3	0.9739	0.9479	0.9307	0.9068	0.8966	0.8628	0.8161	0.7963
4	0.9629	0.9264	0.9022	0.8688	0.8546	0.8079	0.7441	0.7173
6	0.9413	0.8840	0.8466	0.7954	0.7738	0.7036	0.6099	0.5715

μ	w12	w16	box16	w24	w32	box32	w48	w64
1	0.9558	0.9403	0.9335	0.9108	0.8785	0.8639	0.8143	0.7400
1.2	0.9320	0.9080	0.8976	0.8628	0.8133	0.7913	0.7169	0.6083
1.5	0.8969	0.8607	0.8452	0.7932	0.7202	0.6883	0.5825	0.4349
2	0.8399	0.7848	0.7612	0.6837	0.5775	0.5327	0.3895	0.2105
3	0.7314	0.6433	0.6066	0.4891	0.3395	0.2812	0.1196	0.0033
4	0.6305	0.5159	0.4694	0.3270	0.1643	0.1094	0.0046	0
6	0.4510	0.3032	0.2478	0.1004	0.0026	0	0	0

(The exact form depends on the kernel only through $\lambda_{\text{thr}}W_{\text{eff}}$ and $h[0]\sum k$, up to the normalization by U_p , so the $\tau = 512$ value for a kernel reappears at $\tau = 64$ for the kernel with one eighth of its W_{eff} : box32 at $\tau = 512$ read 0.9395/0.8546 at $\mu = 2/4$, which is box4 here.) The bound reaches 0.95 only for $\mu \leq 1.19$ (box8), 1.76 (box4), 2.90 (box2), 5.2 (p0), never for $W \geq 16$; and 0.90 for $\mu \leq 1.76$ (box8), 1.18 (box16), never for box32 or w64. Tightness: the tuned island (support valid, gaps chosen so that the per-sample leverage on the support sits just below λ_{thr} ; $G = 68/125/195 = 1.1/2/3\tau$ for box32) gives 0.784/0.527/0.315 for box32 at $\mu = 2/3/4$ against LB = 0.533/0.281/0.109, and 0.634/0.394/0.371 for w64 against 0.211/0.003/0; the slack is the recovery term, which is now larger than the bound itself; and at $\mu = 4$ nothing is enlarged and A, C and the un-enlarged algorithm coincide. The annealed worst cases for C (Table 2) are $\min(\text{gap} = W \text{ floor}, \text{tuned island})$ to 7×10^{-3} at every μ — the island gains that much from additional far gaps at $1-8\tau$ that lower the recovery term — except for w64 at $\mu = 1.2$, where the 64-gap’s edge leverage ($1.42 c_2/\tau$) makes the rule enlarge its edges, disabling the $\text{gap} = W$, and the worst case is instead a 15-sample gap inside the support plus scattered single samples that push the whole support above threshold (0.789: the n' -term of a fully inpainted support). They are 0.08–0.42 above the bound, which nothing approaches at $\tau = 64$. Under a budget B on the original mask (annealing, as in §A.4), the rule changes nothing for $B \leq 2\tau$ — the worst masks there are the $\text{gap} = W$ and the $B/2$ -island, neither of which is enlarged at $\mu \geq 2$ since a 64-gap’s edge leverage is below threshold — and from $B = 4\tau$ on it holds the island at its tuned isolation: 0.78 ($\mu = 2$) and 0.52 ($\mu = 3$) for box32, 0.63 and 0.39 for w64, against 0.13 and 0.30 without the rule. For algorithm A at $\mu = 2$ a comb-island built from 4τ masked samples is already within 0.045 (box32) and 0.03 (w64) of the unconstrained worst case of Table 2.

A.6 Algorithm A: the b -term

Write $a = M_2k$ and $b = (M - M_2)S_0k$, so $\tilde{k} = a + b$ in (A.5) and

$$V'_A - V'_C = \|R_m(a + b)\|^2 - \|R_ma\|^2 = 2a^T R_m^2 b + \|R_mb\|^2 = -2a^T (2L_m - L_m^2) b + \|R_mb\|^2, \quad (\text{A.11})$$

using $R_m^2 = M - 2L_m + L_m^2$ on the valid subspace and $a^T M b = 0$ (disjoint supports). Near the diagonal $L_m > 0$, so the cross term is negative: the first-pass residuals of nearby valid samples are anti-correlated, $\text{Cov}(r_s, r_{s'}) = -2(L_m)_{ss'} + (L_m^2)_{ss'} < 0$, because the masked fit removed their common local mean. Algorithm A drops the data of the enlarged samples but keeps their imprint on $\hat{f}$, so the second pass re-adds a smoothed copy of their residuals to the kernel’s own residual, with negative covariance. With $b_s \approx h[0]\sum k$ on the enlarged samples within $\sim \ell$ of the kernel, the cross term is of order $-(\sum k)^2 h[0] \times (\text{fraction of the } L_m \text{ row mass on enlarged samples})$, i.e. first order in W_{eff}/τ and independent of μ ; numerically $-0.5 W_{\text{eff}}/\tau$ when the kernel’s run is enlarged on both sides (-0.25 for box32 from $\mu = 1.1$ to 2, -0.21 at $\mu = 3$, -0.13 at $\mu = 4$;

-0.34 for w64 up to $\mu = 2$) and $-0.25 W_{\text{eff}}/\tau$ for one side (-0.12 box32, -0.18 w64, at the new edge of a long gap): independent of μ up to $\mu \approx 2$ and gone by $\mu = 4$, where the rule enlarges nothing beside a long gap. On the same masks V'_C is *above* 1 by 6–19% (the kept run’s residual plus the high-passed n' on its enlarged neighbours), which is the upper-side remark of §A.3; a dedicated maximization finds larger values at small μ , 1.083 / 1.18 at $\mu = 2$ and 1.136 / 1.28 at $\mu = 1.2$ for box32 / w64 (§A.11). Per-sample leverage gives no handle on the b -term: b lives on samples *above* threshold, and its effect is a cross-covariance. Consequences:

- A’s worst case saturates as $\mu \rightarrow 1$ at 0.84 (box32) and 0.79 (w64) at $\mu = 1.2$, i.e. $1 - 0.32$ and $1 - 0.23 W_{\text{eff}}/\tau$, limited by the comb-island below, while C is at the gap-filling floor 0.871/0.789; making the rule more aggressive stops helping below $\mu \approx 1.5$ while its cost keeps growing. A does not cross below the exact-form $\text{LB}(\mu)$ at any $\mu \geq 1.2$ (margin ≥ 0.05 for box32, ≥ 0.18 for w64): the b -term is at full strength on its worst masks, but their a -part $\|R_m a\|^2$ is far from the extremal value the bound assumes, and the bound is much weaker than every found worst case.
- The enlargement is counter-productive for a kernel on the kept run next to an enlarged zone (V'_A is below the un-enlarged value at that position at every μ ; on the comb-island masks below the un-enlarged algorithm gives 0.84/0.94/0.99 against A’s 0.84/0.79/0.70 at $\mu = 1.2/1.5/2$ for box32), while it is what fixes the kernels inside the zone; the net worst case at a long-gap edge still improves (0.628 un-enlarged $\rightarrow$ 0.836/0.899/0.935 at the new edge for $\mu = 2/1.5/1.2$, box32).
- The annealed worst cases for A at $\mu \leq 2$ are all b -term configurations, and all “comb-islands”: the support kept at exactly threshold leverage, surrounded over $1-4\tau$ by an irregular above-threshold comb of 1–6-sample valid runs that the rule enlarges entirely, so that m_2 is a bare island inside a masked neighbourhood (0.701 at $\mu = 2$ for box32, where C on the same mask gives 1.084 and the un-enlarged algorithm 0.99: the b -term, $-0.24 = -0.48 W_{\text{eff}}/\tau$, is the whole effect). Behind it at $\mu = 2$ are a kept run between two long gaps (0.755), the island beside an above-threshold comb (0.731), the tuned island (0.784) and the first kept samples at the new edge of a long gap (0.836). At $\mu = 3$ the box32 optimum is a weak comb-island (0.514, $b = -0.08$) while for w64 it is already the bare island ($b = 0$); at $\mu = 4$ both are bare islands with nothing enlarged (box32: the tuned island, 0.313; w64: a partial-support island keeping the core and one shoulder, 0.360), and A and C coincide. For p0 the comb-island holds up to $\mu = 3$ ($b = -0.009 \approx -0.6 W_{\text{eff}}/\tau$).

What the bound can say for A is only $\|R_m(a + b)\| \geq \|R_m a\| - \|b\|$ with $\|b\|^2 \leq \|S_0 k\|^2$, a loss of order $\sqrt{W_{\text{eff}}}/\tau$, which is useless. For algorithm C the whole question disappears.

A.7 Overmasking cost

Isolated gap of length G . Extra samples masked on *each* side of the gap (where $\lambda > \mu c_2/\tau$), with valid data of 10ℓ on both sides; “free” is a mask extending to the end of the stream:

G	$\mu = 1.05$	1.1	1.2	1.5	2	3	4	6
4	0	0	0	0	0	0	0	0
16	53	0	0	0	0	0	0	0
64	94	72	45	0	0	0	0	0
128	101	83	62	26	0	0	0	0
256	103	87	69	41	15	0	0	0
512	103	88	71	47	25	0	0	0
2048	106	88	72	50	31	10	0	0
free	112	88	72	51	33	13	0	0

The table is scale-invariant: it is the $\tau = 512$ table at $8\times$ the gap, divided by 8. The excess leverage decays over $1-2\tau$ with a long tail (3% above interior at $2.7-3.2\tau$ for $G = 2048$ and a free end), which is why small μ masks so much; nothing is expanded for gaps up to 128 once $\mu \geq 2$ and up to 512 once $\mu \geq 3$, and at $\mu \geq 4$

nothing at all, since even a free end has edge leverage $3.96 c_2/\tau$ (it expands by 6/0/0/1/2/3 samples more than the 2048-gap at $\mu = 1.05$ –3).

Random masks. For Bernoulli masks of valid density n the mean leverage is $c_2 n^{-3/4}/\tau$ to 0.3%, with a per-sample scatter of $\pm 4\%$ ($n = 0.9$) to $\pm 8\%$ ($n = 0.5$) (it was $\pm 1\%$ at $\tau = 512$: the homogenization averages over $\ell = 90$ samples rather than 724), so the rule is a step: everything with $n < \mu^{-4/3}$ is masked (0.94, 0.88, 0.78, 0.58, 0.40, 0.23, 0.16, 0.09 at $\mu = 1.05, 1.1, 1.2, 1.5, 2, 3, 4, 6$), nothing above it, with a transition width of 0.03 ($n = 0.95$) to 0.2 ($n = 0.5$) in μ . Periodic short gaps behave the same way through their density while the period is $\ll \tau$ (e.g. 1 masked of every 8, or 8 of every 64, $n = 0.875$, is masked wholesale at $\mu = 1.1$ and untouched at 1.2); for periods $\geq 2\tau$ only the samples near each gap are above threshold (70/45/32% of the valid samples at $\mu = 1.1$ for 16 of 128, 32 of 256, 64 of 512). For random masks of gaps with log-uniform lengths on a 2^{18} -sample stream, the fraction of valid samples lost is

lengths	cover	$\mu = 1.05$	1.1	1.2	1.5	2	3	4	6
[1, 512]	10%	13.4%	9.7%	6.3%	2.8%	0.87%	0.02%	0.01%	0.01%
[1, 512]	20%	26.9%	20.2%	13.6%	6.4%	2.2%	0.16%	0.07%	0.01%
[1, 4096]	10%	2.6%	2.0%	1.5%	0.84%	0.40%	0.08%	0	0
[1, 4096]	20%	6.1%	4.6%	3.4%	1.9%	0.94%	0.21%	0.01%	0

The [1, 512] distribution is the $\tau = 512$ baseline’s [1, 4096] distribution scaled by 1/8 and reproduces its losses; with lengths up to 4096 the gaps are mostly $\gg \tau$, with edges already at $\sim 3.6 c_2/\tau$, and the loss in fraction is smaller.

A.8 Numerical methods and validation

Exact evaluation. For a given mask, (A.5) needs $\mathcal{N}_m^{-1}$ applied to a few vectors, and $\lambda[t]$ needs its diagonal. $\mathcal{N}_m$ is pentadiagonal but ill-conditioned: its largest eigenvalue is 16ρ and its smallest is set by the smoothest modes, giving a condition number of 3×10^8 for benign masks and up to 10^{13} for a free end (the free modes of a long masked tail) at $\tau = 64$, against 10^{12} – 10^{16} at $\tau = 512$. Dense algebra ($T \leq 4521$) is adequate at $\tau = 64$ except for the free-end masks, where a dense LU loses $\sim 10^{-5}$ in the masked tail; a plain banded Cholesky loses the smooth modes, three steps of iterative refinement recover them, and the refined banded solver agrees with an information-form two-filter smoother (the forward and backward recursions of *detrending.pdf*, “Time detrending algorithm 2: Kalman filter”, run over the whole timestream) to 9×10^{-14} on every mask, including the free ends (and to 10^{-6} with 30000-sample gaps at $\tau = 512$). The banded solver with refinement was used everywhere (annealing, exact tables, Monte Carlo), the information-form smoother for the leverage diagonal $\lambda[t] = ((J_f + J_b)^{-1})_{00}$ and as the cross-check of every annealed optimum (agreement 10^{-6}), and dense algebra for the eigenvalues and the solver cross-checks.

Adversarial search. Simulated annealing on $\pm \log(V'/U)$ over masks in a window of half-width 24τ around the kernel (valid clamps of 8ℓ beyond), with single-sample flips, masking or unmasking of random intervals, count-preserving transfers, boundary shifts and imposed periodic patterns, optionally under a budget on the number of masked samples, followed by a deterministic single-flip descent. For algorithm 0, 7–15 seeds per problem including the analytic candidates, 60000–90000 steps, 353 runs; for A and C, with the rule inside the loop, 11 seeds per problem (including the algorithm-0 worst masks), 40000 steps plus a two-pass descent over the whole window, 776 runs over the unconstrained and budgeted problems. Everything was run natively at $\tau = 64$ with the real kernels. At fixed W/τ every length in the problem scales with τ except the island isolation length ($\tau^{4/3}$), so a search at another τ with the same W/τ would transfer; but the problem depends on W/τ , and nothing was transplanted from the $\tau = 512$ baseline ($W/\tau = 1/16$) or from narrower kernels. Every reported optimum is locally optimal under all single flips. Annealing cannot certify a global optimum; the evidence is the agreement of independent seeds and the fact that no run ever went below the best analytic candidate by more than the amounts stated in §A.4 and §A.5.

Monte Carlo. The literal two-pass procedure (masked smoother, inpainting, unmasked smoother, then the 16 convolutions; nothing uses the identity or the variance formulas) was simulated at $\tau = 64$ with

10^5 – 6×10^5 noise realizations per mask for eleven masks covering every family above (gap = W , free and finite long-gap edges, the $G_{1/2}$ island, the adversarial island, a sparse periodic mask, the comb, and the two annealed maximizing masks), all 16 kernels at every position of a ± 256 scan, giving 143632 (kernel, position) comparisons with the exact formulas: pull r.m.s. 0.93–1.05, maximum $|\text{pull}|$ 4.7, 443 beyond 3σ against 388 expected (the two short-gap masks have slightly heavy tails), mean pull -0.05 from the shared realizations. The comb excess above 1 was resolved at 61σ (box32: 1.02100 ± 0.00034) and 89σ (w64), and for every kernel of width ≥ 2 , while p0 is below 1 (0.99752 ± 0.00010); the annealed maxima were confirmed at 12σ (box32) and 25σ (w64) above 1. For the enlarged algorithms the literal three-pass procedure was simulated at $\tau = 64$ on the four worst masks at $\mu = 2$ (2×10^5 realizations each; e.g. box32: 0.6996 ± 0.0022 vs 0.7009 for A, 0.7797 ± 0.0024 vs 0.7805 for C; w64: 0.5934 ± 0.0018 vs 0.5943 and 0.6289 ± 0.0020 vs 0.6293), 128 (kernel, variant) comparisons with pull r.m.s. 0.87 and maximum 2.2; among them $V'_C/U = 1.040$ for w64 on the box32 comb-island mask, C above 1 at 12σ .

A.9 Remarks: algorithm B, fixed lag, renormalization

Algorithm B (re-run the first pass with m_2) has no b -term but its kept samples carry the residual of the *smaller* mask, whose leverage is higher near the new edges and on any short kept run. At the new edge of a long gap it reproduces the un-enlarged edge value (0.628 for box32) at every μ , because masking the edge just moves the $4\times$ edge leverage to the new edge; on a kept run between two enlarged zones it turns the run into an island, at or near the island floors of §A.4. Iterating the rule to self-consistency does not help: for μ below the long-gap edge leverage (3.6–4) each iteration creates a new above-threshold edge and the expansion eats the whole timestream.

Fixed lag. With the production fixed-lag estimator ($L = 4\ell = 363$ at $\tau = 64$) the identities (1.7) no longer hold exactly, since neither pass is a whole-timestream solve. An exact fixed-lag computation (an information-form engine validated against the python reference `KalmanDetrender`) shows that for algorithm 0 nothing moves by more than 3×10^{-3} : the free-edge values are reproduced to 2×10^{-4} , the 32-island at $G_{1/2}$ (gaps $117 \ll L$, with 2.7ℓ of valid data inside the lookahead beyond them) moves from 0.560 to 0.558, and a finite gap longer than L ($G = 2048$) behaves as a free gap (0.599 instead of 0.628), because the far data no longer clamp the baseline on the lookahead side. For algorithms A and C the rule is sensitive to which leverage it uses: on the whole-stream worst masks at $\mu = 2$, tuned to the threshold to 10^{-3} , the fixed-lag leverage on the support is 0.1–4.6% higher (2.07–2.09 against $2.00 c_2/\tau$ on the comb-islands), so the fixed-lag rule drops part or all of the support and the values jump (the box32 comb-island: A from 0.70 to 1.006, with the whole support inpainted). These are not fixed-lag worst cases — an adversary tuning to the fixed-lag leverage would rebuild the same structures 1–5% more isolated. The conclusions of this section are qualitative in that setting; the bound (A.9) is not.

Renormalization. $V'_p(t)$ is a deterministic function of the mask, so the principled route to a guarantee near 1 is not to mask more but to compute it and divide: the kernel-level $1 - 2\lambda_k + \nu_k$ needs the W -band of $\mathcal{N}_m^{-1}$ and of $\mathcal{N}_m^{-1} M \mathcal{N}_m^{-1}$ (the ν of *detrending.pdf*, “Outputs” extended to lags up to W), roughly $O(W)$ 2×2 operations per sample on top of the filter. This has not been pursued.

A.10 Dependence on τ

The problem depends on the kernel and the smoother only through W_{eff}/τ (and A/τ for an island of A valid samples). At fixed W/τ every length scales with τ — the edge-leverage profiles, the expansion radii of §A.7 and the comb structures at $\tau = 64$ are the $\tau = 512$ ones at one eighth of the distance, and the fitted island constant c of (A.8) is the same function of G/τ — except the island isolation length, which scales as $\tau^{4/3} A^{-1/3}$ ($G_{1/2} = 647$ for $A = 1$ at $\tau = 64$ against 11048 at $\tau = 512$, a factor $17 = 8^{4/3}$). The previous baseline of these notes was $\tau = 512$ with the same 16 kernels ($W_{\text{eff}}/\tau = 1/16$ for box32), where the headline numbers were: plain-inpainting worst cases 0.984 (kernel filling a gap of its own width) and 0.929 (kernel beside a long gap) for box32, floors 0.017 (box32) and 0.135 (w64) reached at isolation gaps of ~ 2900 samples; worst found for A and C at $\mu = 2$ of 0.957 and 0.975 (box32); largest excess above 1

of 1.0027 (box32). The $\tau = 64$ deficits are not those numbers times 8: for box32 the valid-side edge deficit goes from 0.071 to 0.401 (5.6 $\times$), the gap = W deficit from 0.0165 to 0.129 (7.8 $\times$) and the comb excess from 0.0023 to 0.021 (9 $\times$), because the leading-order regime, in which every deficit is a fixed multiple of W_{eff}/τ , ends around $W_{\text{eff}}/\tau \approx 0.15$ (box8 at $\tau = 64$ is still inside it: the linear edge formula gives 0.863 against 0.865 exact). In units of W_{eff}/τ the worst-case deficit coefficients of Table 2 survive from $\tau = 512$ to $\tau = 64$ for box32 to 10–20% (for C, exactly), but not for w64, whose deficit saturates at its core-island floor; and the exact-form bounds of §A.3 and §A.5, which depend on the kernel only through $\lambda_{\text{thr}}W_{\text{eff}}$ and $h[0]\sum k$, transfer between a kernel at one τ and the kernel with W_{eff} scaled at the other.

A.11 The windowed figure of merit

This subsection contains the details behind §1.4: the evaluation of the window average, the searches for X_{hi} and $\tilde{X}_{\text{lo}}$, the full per-kernel tables, the T_c dependence, the Monte Carlo checks and the caveats. Everything is at $\tau = 64$ with $T_c = 1024$, in the setting of §1.1. Labels are as in §1.3: “exact” means exact linear algebra for the mask named; “exact scan” the maximum over a finite family of masks, each evaluated exactly; “annealing” or “best found” a value found by simulated annealing, exact for the mask exhibited but not certified as the maximum; “Monte Carlo” a simulation of the literal procedure.

The windowed evaluator. $\tilde{X}_{\text{lo}}$ needs $V'_p(t)$ at every position of a window $[t_0, t_0 + T_c)$ for every kernel, 16×1024 quadratic forms per mask. For algorithm 0, let $S = [t_0 - 32, t_0 + T_c + 32)$ be the samples any kernel in the window touches and $Z = \mathcal{N}_m^{-1}ME_S$ (one banded solve with $|S|$ right-hand sides; the masked columns are zero and are skipped). The S -block of $R_m^2 = M - 2L_m + L_m^2$ is $M_S - 2\text{diag}(m_S)Z_S + Z^T M Z$, of which only the band $|s - s'| < W_{\text{max}}$ is needed, since it is contracted with kk^T ; the band of $Z^T M Z$ is computed blockwise (cost $T \times |S| \times 128$ instead of $T \times |S|^2$), and $\|R_m k_{p,t}\|^2$ at every t is a strided quadratic form on the band. The n' -term $\|(I - M)(k - S_0 k)\|^2$ is the correlation of $1 - m$ with the squared impulse response of $(I - S_0)$ applied to k (kept to ± 2560 samples, $e^{-2560/\ell} \sim 5 \times 10^{-13}$). For A and C the vectors $M_2 k_{p,t}$ and, for A, $(M - M_2)S_0 k_{p,t}$ are formed for every t in the window and solved in one banded solve with T_c right-hand sides per (kernel, variant), using $\|R_m v\|^2 = \|v\|^2 - 2v^T w + \sum_s m_s w_s^2$ with $w = \mathcal{N}_m^{-1}v$ for v supported on valid samples; the leverage $\lambda[t]$ that the rule needs at every t comes from the banded Cholesky factor by the Takahashi recursion (the band of $\mathcal{N}_m^{-1}$ from the factor, $O(T)$). The banded Cholesky with iterative refinement of §A.8 is used throughout; in the clamped geometry below the refinement changes nothing (4×10^{-10}). Cost at $T = 3072$: 0.1–0.2 s per mask for all 16 kernels (algorithm 0), 0.1 s per (mask, kernel, variant) for A and C. A pointwise evaluator (one or two right-hand sides, ~ 1 ms) drives the X_{hi} annealing.

Validation. The evaluator reproduces every reference number of §A.1–§A.7 to the digits given there: the constants of the smoother, U_p (Table 1), the gap = W , free-edge, finite-gap-edge, island, comb and sparse-periodic values of §A.4 (e.g. the two-sided island scan 0.930...0.129, including 0.560 at $G = 117$), $B_{1,p}/U_p$, the isolated-gap expansion table of §A.7 entry by entry, the density threshold, the tuned islands 0.784/0.527/0.315 and 0.634/0.394/0.371 of §A.5, the new-edge values 0.835/0.900/0.936 of §A.6, and Table 2 to 0.001–0.006 for 28 of its 30 entries (the two C entries at $\mu = 1.2$ were seed-limited by 0.005/0.015 in the re-run). Against dense algebra (dense $\mathcal{N}_m^{-1}$, R_m and a finite- T S_0 at $T = 2048$ on structured random masks) the windowed evaluator agrees to 1.2×10^{-8} (algorithm 0, 16 kernels at 29 positions; the dense solve’s own rounding at $\kappa \sim 10^8$), 2.4×10^{-7} (A) and 4.6×10^{-8} (C), the Takahashi leverage to 8×10^{-9} , and the pointwise and windowed evaluators agree to 2×10^{-10} ; $m \equiv 1$ gives $V' = U$ at every position to 6×10^{-11} , $m \equiv 0$ to 8×10^{-15} , and at $\mu = \infty$ A = C = algorithm 0 to 4×10^{-10} . Monte Carlo of the literal procedure on one structured mask, 32000 realizations, all 16 kernels at all 1024 window positions: pull r.m.s. 1.011 and maximum |pull| 3.9 over 16384 (kernel, position) pairs for algorithm 0, and 1.02 / 4.65 for each of A and C at $\mu = 2$.

Geometry. For $\tilde{X}_{\text{lo}}$ the window is $[1024, 2048)$ of a $T = 3072$ stream; the mask is free on the window plus $8\tau = 512$ margins on each side (2048 free samples) and all valid beyond (clamps of 512 against 2048 change the window average by 6×10^{-10}). The rule is applied to the whole stream, so its ends expand as in §A.7; they are $\geq 11\ell$ from the window. For X_{hi} the mask is free within $\pm 24\tau$ of the kernel for algorithm 0 (as in

§A.8) and within $\pm 12\tau$ for A and C (the maximizing structures live within $\sim 6\tau$, and an enlarged sample beyond 8τ enters only through the b -term, at $< 3 \times 10^{-3}$ relative), with valid clamps of 8ℓ beyond.

Algorithm 0: exact scan over periodic masks. Valid runs of length A and gaps of length B (the sparse periodic masks of §A.4), with A in $\{1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 96, 128, 192, 256\}$ and B in $\{1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 40, 48, 56, 64, 80, 96, 112, 128, 160, 192, 224, 256, 320, 384, 448, 512, 640, 768\}$ (448 pairs), every mask evaluated exactly for all 16 kernels in two variants: *periodic* over the whole stream, with the window average maximized over the phase exactly (a sliding window of T_c over V' on $T_c + P$ positions), and *clamped*, periodic only over the window and its margins with valid data beyond (the search space of the annealing), at 8 phases. The two agree to 10^{-4} for every kernel. Table 4 gives the maximum over the grid (label: exact scan) with the worst (A, B), the best member of three sub-families — the island of the kernel’s own width, the kernel filling a gap of its own width, and the dense combs ($A, B \leq 8$) — and the pointwise deficit $1 - \min_t V'/U$ on the worst mask, which shows what the average does to it.

kernel	worst (A, B)	$\tilde{X}_{lo}$	island $A = W$	gap = W	comb	$1 - \min_t V'/U$
p0	(16, 128)	0.0035	(1, 6) 0.0033	(1, 1) 0.0024	(1, 6) 0.0033	0.025
box2	(16, 128)	0.0068	(2, 16) 0.0045	(3, 2) 0.0019	(4, 8) 0.0047	0.051
w3	(16, 128)	0.0090	(3, 32) 0.0061	(4, 3) 0.0025	(4, 8) 0.0054	0.068
w4	(16, 128)	0.0119	(4, 40) 0.0082	(6, 4) 0.0032	(4, 8) 0.0058	0.092
box4	(16, 128)	0.0130	(4, 56) 0.0084	(6, 4) 0.0028	(4, 8) 0.0054	0.103
w6	(32, 160)	0.0168	(6, 96) 0.0120	(8, 6) 0.0044	(4, 8) 0.0043	0.082
w8	(32, 160)	0.0224	(8, 160) 0.0169	(12, 8) 0.0061	(4, 8) 0.0016	0.112
box8	(32, 160)	0.0246	(8, 160) 0.0174	(16, 8) 0.0052	(4, 8) -0.0007	0.126
w12	(32, 160)	0.0320	(12, 112) 0.0254	(16, 12) 0.0088	< 0	0.170
w16	(32, 160)	0.0419	(16, 128) 0.0359	(24, 16) 0.0126	< 0	0.235
box16	(32, 160)	0.0452	(16, 128) 0.0366	(32, 16) 0.0110	< 0	0.264
w24	(32, 160)	0.0568	(24, 224) 0.0522	(32, 24) 0.0193	< 0	0.366
w32	(48, 192)	0.0715	(32, 160) 0.0703	(48, 32) 0.0285	< 0	0.350
box32	(48, 192)	0.0758	(32, 160) 0.0715	(48, 32) 0.0253	< 0	0.399
w48	(48, 192)	0.0931	(48, 192) 0.0931	(64, 48) 0.0453	< 0	0.576
w64	(64, 128)	0.1113	(64, 128) 0.1113	(64, 64) 0.0641	< 0	0.440

Table 4: Algorithm 0, exact scan over periodic masks (valid runs A , gaps B) at $T_c = 1024$: the maximum of the window deficit over the grid and the mask attaining it; the best member of the island $A = W$, gap = W and dense-comb ($A, B \leq 8$) sub-families, as (A, B) and value; and the pointwise deficit on the worst mask. Each entry is exact for its mask, with the phase maximized exactly.

The worst periodic family is the sparse island: valid runs of about 1–1.5 W (for $W \geq 16$; 16–32 samples for the narrower kernels) separated by gaps of 2–3 τ , valid fraction ~ 0.2 . The landscape is smooth and the optimum broad (box32 is within 0.004 of its maximum over $A = 32$ –64, $B = 96$ –224), and the phase matters at the 0.003–0.009 level for periods of 200–300 (T_c/P not an integer). The gap = W family, the pointwise worst case at small budgets (0.871 for box32), is $3\times$ weaker in the average, because only one position per gap is at the minimum and the masked positions around it are restored to ~ 1 ; dense combs give a negative deficit (the comb excess of §A.3); and for the tapered kernels the island of the full support beats the core-only island in the average (w64: 0.1113 against ≤ 0.084), the opposite of the pointwise ordering of §A.4. The last column is the point: on the worst periodic mask the pointwise minimum is 0.40 (box32) and 0.44 (w64) while the average deficit is 0.076 and 0.111. The finer grid used for A and C below raises the algorithm-0 maximum by up to 0.003 for the wide kernels (box32 0.0785 at (48, 150), w48 0.0952 at (48, 150), w64 0.1146 at (64, 177)).

Algorithm 0: annealing. Simulated annealing on the window average with the moves of §A.8 (single flips 45%; masking or unmasking of an interval of log-uniform length 1–512, 20%; an imposed periodic pattern of period 2–768 and duty cycle 0.05–0.95 on a random interval, 20%; run-boundary shifts of 1–16 samples, 15%), a geometric temperature schedule from 2×10^{-3} to 2×10^{-6} , 4000 steps (each one multi-right-hand-side solve) plus a single-flip descent at 600 random positions; six kernels, six seeds each (the best

and second-best periodic masks of the clamped scan at the phase of their minimum, the best gap = W and island $A = W$ patterns, Bernoulli(0.5), and random log-uniform gaps covering 30%). Label: annealing (best found).

kernel	annealing	6 seeds	exact scan	islands in the window	gaps
p0	0.0040	0.0039–0.0040	0.0035	single samples (runs 1–11, median 1)	1–60, median 3–4
box8	0.0281	0.0267–0.0281	0.0246	$9 \times (19\text{--}29)$, $\sim 3W$	85–131
box16	0.0505	0.0488–0.0505	0.0452	$7 \times (31\text{--}37)$, $\sim 2W$	114–145
box32	0.0858	0.0816–0.0858	0.0758	$6 \times (41\text{--}50)$, $\sim 1.4W$	111–167
w48	0.1046	0.0982–0.1046	0.0931	$6 \times (47\text{--}55)$, $\sim 1.1W$	127–154
w64	0.1225	0.1203–0.1225	0.1113	$6 \times (53\text{--}59)$, core + part of the shoulders	107–136

The annealer converges to the scan’s structure from every seed (the Bernoulli and random-gap seeds start at negative or tiny deficits) and improves it by 10–15% in two ways: it masks the margins entirely (gaps of 512–570 flanking the window, so that the kernels near the window edges also see a free-edge deficit and the islands lose the clamps’ support), and it detunes the periods and gap lengths slightly (each island’s isolation against its neighbours; p0 ends as an irregular sparse comb at valid fraction 0.09–0.16). The seeds are within 0.002–0.006 of each other, i.e. not fully converged at 4000 steps; a longer run would probably add $\sim 0.002\text{--}0.005$. The best box32 mask, as run lengths (m = masked, V = valid): m517 V41 m158 V48 m167 V50 m143 V46 m167 V47 m111 V41 m512.

Comparison with X_{lo} , and the T_c dependence. Against the pointwise X_{lo} of algorithm 0 — the unbounded island floors of §A.4, 0.996, 0.967, 0.934, 0.871, 0.683, 0.705 for p0, box8, box16, box32, w48, w64 (core-only for w64) — the annealed $\tilde{X}_{\text{lo}}$ is 0.004, 0.03, 0.05, 0.10, 0.15, 0.17 of it. (The island floors of the other kernels, computed the same way — support valid, gaps of 2048 on each side, core-only for the tapered kernels — are 0.991 (box2), 0.875 (w3), 0.882 (w4), 0.983 (box4), 0.862 (w6), 0.865 (w8), 0.836 (w12), 0.831 (w16), 0.785 (w24), 0.762 (w32); these are the best analytic candidates for X_{lo} , not annealed.) The pointwise adversary drives one kernel position to 0.13–0.30 with ~ 2000 masked samples around an island; in the average those samples are restored to ~ 1 , and the optimum is $\sim 20\%$ valid at a per-island deficit of 0.4–0.6, i.e. 0.08–0.12. Table 5 gives the exact, phase-maximized window average of the fully periodic masks that are worst at $T_c = 1024$ and at $T_c = 256$, for $T_c = 256, 1024, 4096$ and $T_c \rightarrow \infty$ (the period average).

kernel	(A, B)	$T_c = 256$	1024	4096	∞	$1 - \min_t V'/U$
p0	(16, 128)	0.0035	0.0035	0.0032	0.0032	0.025
p0	(16, 224)	0.0051	0.0033	0.0030	0.0028	0.039
box8	(32, 160)	0.0312	0.0246	0.0229	0.0224	0.126
box8	(32, 192)	0.0342	0.0231	0.0221	0.0214	0.143
box16	(32, 160)	0.0570	0.0452	0.0422	0.0413	0.264
box16	(32, 192)	0.0627	0.0427	0.0408	0.0396	0.300
box32	(48, 192)	0.0885	0.0758	0.0693	0.0680	0.399
box32	(128, 768)	0.1009	0.0467	0.0331	0.0302	0.389
w48	(48, 192)	0.1085	0.0931	0.0852	0.0836	0.576
w48	(128, 768)	0.1297	0.0603	0.0428	0.0391	0.399
w64	(64, 128)	0.1307	0.1113	0.1064	0.1048	0.440
w64	(128, 768)	0.1669	0.0776	0.0557	0.0509	0.427

Table 5: Algorithm 0: dependence of the window deficit on the chunk length T_c , for the periodic mask worst at $T_c = 1024$ (first row of each kernel) and the one worst at $T_c = 256$ (second row); exact, phase maximized; $T_c = \infty$ is the period average. The last column is the pointwise deficit on the mask.

The periodic-family maximum goes $0.101 \rightarrow 0.076 \rightarrow 0.069 \rightarrow 0.068$ (box32) and $0.167 \rightarrow 0.111 \rightarrow 0.106 \rightarrow 0.105$ (w64) for $T_c = 256 \rightarrow 1024 \rightarrow 4096 \rightarrow \infty$: at $T_c = 1024$ the value is within $\sim 10\%$ of the limit, while $T_c = 256$ (4τ) lets the adversary use a single large island with long gaps ($A = 128$, $B = 768$: a 4τ window sitting on 2τ of valid data flanked by 12τ gaps). For the annealed masks the maximum over

256-sample sub-windows of the 1024 window is 0.0049 / 0.0358 / 0.0606 / 0.1139 / 0.1318 / 0.1580 (p0 / box8 / box16 / box32 / w48 / w64), and tiling the 2048-sample free pattern with period 2048 gives a diluted 0.0025 / 0.0148 / 0.0286 / 0.0460 / 0.0563 / 0.0659 at $T_c = 4096$ (not re-optimized: at $T_c = 4096$ the adversary would use the periodic values of the table). So the reduction from X_{10} to $\tilde{X}_{10}$ is the averaging itself, not the particular chunk length; T_c enters through the boundary term (kernels near the window edges beside long gaps, $O(\tau/T_c)$), which is what the annealer exploits at $T_c = 1024$ for its 10–15% over the periodic masks.

Algorithms A and C: exact scan. The same grid plus, for each μ , 14 run lengths $\times$ 12 densities around $\mu^{-4/3}$ (900 pairs in all); the mask is periodic over the whole stream, the rule is applied to it (so V' is periodic and one period is evaluated), and the window average is maximized over the phase exactly. Table 6 gives the maximum over the grid for all 16 kernels (label: exact scan).

kernel	alg. 0	algorithm A					algorithm C				
		$\mu = 1.2$	1.5	2	3	4	$\mu = 1.2$	1.5	2	3	4
p0	0.0035	0.0015	0.0020	0.0028	0.0033	0.0034	0.0011	0.0020	0.0028	0.0033	0.0034
box2	0.0068	0.0029	0.0039	0.0054	0.0065	0.0067	0.0020	0.0039	0.0054	0.0065	0.0067
w3	0.0090	0.0039	0.0052	0.0072	0.0085	0.0088	0.0026	0.0052	0.0072	0.0085	0.0088
w4	0.0119	0.0052	0.0069	0.0096	0.0114	0.0118	0.0035	0.0069	0.0096	0.0114	0.0118
box4	0.0130	0.0057	0.0076	0.0106	0.0126	0.0130	0.0038	0.0076	0.0106	0.0126	0.0130
w6	0.0168	0.0076	0.0100	0.0140	0.0166	0.0170	0.0050	0.0099	0.0140	0.0166	0.0170
w8	0.0224	0.0101	0.0133	0.0187	0.0220	0.0227	0.0065	0.0130	0.0187	0.0220	0.0227
box8	0.0246	0.0111	0.0146	0.0205	0.0241	0.0248	0.0070	0.0141	0.0205	0.0241	0.0248
w12	0.0320	0.0144	0.0190	0.0268	0.0313	0.0323	0.0089	0.0179	0.0268	0.0313	0.0323
w16	0.0419	0.0188	0.0248	0.0351	0.0407	0.0422	0.0111	0.0233	0.0351	0.0407	0.0422
box16	0.0452	0.0203	0.0266	0.0383	0.0441	0.0455	0.0115	0.0254	0.0383	0.0441	0.0455
w24	0.0568	0.0255	0.0340	0.0496	0.0571	0.0572	0.0131	0.0325	0.0496	0.0571	0.0572
w32	0.0715	0.0320	0.0441	0.0645	0.0741	0.0741	0.0137	0.0415	0.0645	0.0741	0.0741
box32	0.0758	0.0338	0.0469	0.0684	0.0785	0.0785	0.0125	0.0435	0.0684	0.0785	0.0785
w48	0.0931	0.0407	0.0575	0.0838	0.0952	0.0952	0.0117	0.0515	0.0838	0.0952	0.0952
w64	0.1113	0.0464	0.0677	0.0987	0.1146	0.1146	0.0080	0.0579	0.0987	0.1146	0.1146

Table 6: Algorithms A and C, exact scan over periodic masks with the rule applied, $T_c = 1024$: the maximum of the window deficit over the grid (900 (A, B) pairs per μ), exact for every mask with the phase maximized exactly. The algorithm-0 column is the 448-pair grid of Table 4; where the worst mask is not enlarged (all of $\mu = 3$ and 4 for $W \geq 2$) the A and C entries are algorithm-0 values too, and exceed that column by up to 0.003 where the finer grid found a slightly worse mask.

The worst (A, B) are nearly independent of the kernel. At $\mu = 2$ they are (16, 24) for p0, (48, 73) up to w12 and (64, 97) from w16 on, valid fraction 0.40, for A and C alike; at $\mu = 3$ and 4, (32, 106) to (64, 177), valid fraction 0.16–0.27; at $\mu = 1.5$, (48, 34) to (96, 69) for A and (48, 34) to (128, 72) for C, valid fraction 0.58–0.64; at $\mu = 1.2$, (96, 26) and (128, 35) for A (valid 0.79) against (48, 13) to (96, 24) and, for box32, w48 and w64, the gap = W pattern (192, 32) for C (valid 0.79–0.86). Three things follow. (i) The rule pushes the periodic adversary to the density threshold: everything with valid density below $\mu^{-4/3}$ (0.78 / 0.58 / 0.40 / 0.23 / 0.16) is enlarged wholesale and comes back at $V' = U$ (C) or above it (A), so the worst masks sit at density just above $\mu^{-4/3}$ with nothing enlarged (box32 at $\mu = 2$: (64, 97), $n = 0.398$ against 0.397), where $A = C =$ algorithm 0 exactly. At $\mu = 4$ the algorithm-0 optimum ($n = 0.20$ –0.33) is untouched; at $\mu = 3$ the sparse islands at $n = 0.2$ are removed but the (48, 150) island at $n = 0.24$ is nearly as bad; at $\mu = 2$ the density must be ≥ 0.40 and the deficit drops to 0.068 (box32) / 0.099 (w64); at $\mu = 1.5$ to 0.044–0.047 / 0.058–0.068; at $\mu = 1.2$ to 0.013–0.034 / 0.008–0.046. (ii) A’s b -term matters only at $\mu \leq 1.5$, where the rule can enlarge part of a run: the worst A masks there have fewer kept than valid samples (box32 at $\mu = 1.2$: (128, 35), valid 0.79, kept 0.49; at $\mu = 1.5$: (96, 69), valid 0.58, kept 0.44) — the rule drops the edges of every run, A re-adds their smoothed residuals with negative covariance on the kept centre, and A’s window deficit is $2.7\times$ ($\mu = 1.2$) and $1.08\times$ ($\mu = 1.5$) C’s. (iii) Wholesale enlargement of a dense comb gives

a *negative* deficit for A (e.g. (32, 160) at $\mu \leq 3$: -0.004 , while C is 0 exactly): the comb excess reappears as an overshoot, which is X_{hi} ’s business.

Algorithms A and C: annealing. The same objective with the rule inside the loop. Seed candidates per (variant, μ , kernel): the best three periodic masks of the scan in the clamped geometry (four phases each, with and without the margins masked), algorithm 0’s annealed worst mask, and a Bernoulli mask at density $1.1\mu^{-4/3}$; the best two candidates were annealed for 1200 steps plus 100 descent flips (six kernels, five μ , two variants). Label: annealing (best found). The results are the $\tilde{X}_{\text{lo}}$ block of Table 7. The annealer adds only 0.000–0.005 to the scan (box32: 0.0688 against 0.0684 at $\mu = 2$, 0.0349 against 0.0338 at $\mu = 1.2$), against the +0.010 it added for algorithm 0: the margin-masking trick is disabled by the rule (the first kept samples beside a long gap are dropped, 31 per side at $\mu = 2$), and “margins masked” seeds are 0.01–0.02 worse than the periodic mask with valid margins for $\mu \leq 3$. At $\mu = 4$, where nothing beside a long gap is enlarged, algorithm 0’s worst masks are recovered for w48 and w64 (0.1046, 0.1225) and nearly for box32 (0.0826–0.0831 against 0.0858 in 1200 steps: algorithm 0’s box32 mask is partly enlarged at $\mu = 4$, its 41-sample islands with 158–167 gaps sitting just above $4c_2/\tau$, and gives 0.0779 / 0.0738 for A / C).

Structures. For $\mu \geq 2$ the worst windows contain no enlarged samples (in every case but A, box32, $\mu = 2$, with 3 of 439), and A = C = algorithm 0 on them to ≤ 0.002 : the adversary sidesteps the rule by keeping the density just above $\mu^{-4/3}$, at the price of a 3–5 $\times$ smaller per-island deficit than algorithm 0’s sparse islands (box32 at $\mu = 2$: the per-island deficit falls from 0.6 to 0.16 while the valid fraction rises from 0.24 to 0.40; the pointwise minima on the worst masks are 0.81–0.91 for $\mu \leq 2$ and 0.43–0.67 for $\mu \geq 3$). What is enlarged lies in the margins (204 of 375 margin samples for A, box32, $\mu = 2$), which for C are simply masked and for A add a b -term at the window edges (on that mask A = 0.9312, C = 0.9332, algorithm 0 = 0.9318). For $\mu \leq 1.5$ partial enlargement of run edges is possible and the b -term separates A from C: the (128, 35) family at $\mu = 1.2$ has 268 of 817 window samples enlarged (the edges of every 128-run, leaving 1–2-sample gaps in the runs), with A = 0.965 against C = 1.0016 and algorithm 0 = 0.982 on the same mask — the b -term costs A 0.035 in the average, while C’s overshoot beside the enlarged edges lifts its average above 1. Because the overshooting positions beside every enlarged zone are part of the average, alternating zones and kept runs cost A only 0.035 and not the pointwise 0.16–0.30 of §A.6. C’s overshoot never helps the adversary on the lower side (it can only raise the average); C’s worst masks at $\mu = 1.2$ are the gap = W family (box32, 0.013) and, for w64, a (192, 32) pattern whose window is entirely enlarged ($V'_C = U$ there), with a deficit only from the kernels at the window edges (0.009).

The upper side, X_{hi} . Pointwise annealing on $V'_p(t)/U_p$. Algorithm 0: four seeds per kernel (all valid; Bernoulli(0.5) within $\pm 3\tau$; a comb of density 0.4 within $\pm 2\tau$; the comb plus a masked annulus at $2\text{--}6\tau$), 60000 steps plus two full single-flip descent passes; the seeds agree to 6×10^{-4} for every kernel and reproduce the 7-seed optima of §A.3 (1.0247, 1.0504, 1.00027). All 16 kernels (annealing, best found):

	p0	box2	w3	w4	box4	w6	w8	box8
X_{hi}	0.0003	0.0013	0.0017	0.0024	0.0026	0.0036	0.0048	0.0054
	w12	w16	box16	w24	w32	box32	w48	w64
X_{hi}	0.0073	0.0101	0.0114	0.0156	0.0219	0.0247	0.0349	0.0504

i.e. $X_{\text{hi}} \approx 0.050\text{--}0.056 W_{\text{eff}}/\tau$ throughout (0.049 box32, 0.056 w64, 0.043 box8), with the structure of §A.3: a sparse irregular comb of 1–3-sample valid runs at valid fraction 0.2–0.35 on the support and out to τ , ~ 0.7 in the $1\text{--}2\tau$ shell, a fully masked annulus from 2τ to $\sim 6\tau$, a valid patch of $\sim \tau$ at $6\text{--}8\tau$ and essentially masked beyond (for p0, the support and its $\pm 1.5\tau$ neighbourhood valid and the annulus alone).

For A and C: five seeds (all valid; algorithm 0’s maximizer; the kernel on a kept run between two above-threshold combs; the first kept samples at the new edge of a long gap; a comb of density $1.1\mu^{-4/3}$ with the support valid), 20000 steps plus two descent passes, rule inside the loop; the results are the X_{hi} block of Table 7. *Algorithm A never overshoots more than algorithm 0.* The seeds agree to ≤ 0.003 , and the maximizer is algorithm 0’s comb plus annulus with the comb thinned to stay below threshold (11–19 of the 32 support samples valid for box32) and nothing enlarged within τ of the kernel: the b -term, a negative

cross-covariance, can only lower the variance wherever the rule enlarges something near the kernel, so X_{hi} rises monotonically with μ to algorithm 0's value ($0.0205 \rightarrow 0.0246$ for box32, $0.0418 \rightarrow 0.0502$ for w64). *Algorithm C overshoots by much more than the incidental values of §A.6:* 0.136 / 0.111 / 0.083 / 0.057 / 0.043 for box32 and 0.280 / 0.218 / 0.180 / 0.109 / 0.087 for w64 at $\mu = 1.2 / 1.5 / 2 / 3 / 4$ (box16 0.066–0.020, box8 0.032–0.010, p0 0.003–0.0003). The structure is the kernel on a short kept run — box32: 7–22 of the 32 support samples valid and kept; w64 at $\mu \leq 2$: the whole support valid but only 20–32 of 64 kept, the rest enlarged — with everything else within $3\text{--}4\tau$ enlarged (above-threshold combs) or masked: the kept samples return their first-pass residual, the enlarged support samples get fresh noise at nearly full variance, and the negative covariance between the kept and the dropped residuals, which holds algorithm 0 (and A, through the b -term) down, is removed. On the same masks A is at 0.93–0.99 and algorithm 0 at 0.75–1.03; the 1.084 of §A.6 (C on A's comb-island at $\mu = 2$) is a special case, and the 1.18 of w64 at $\mu = 2$ is this maximum. The C seeds spread by 0.01–0.04 for w48 and w64, so those entries are lower bounds; at $\mu = 1.2$ the values 1.136 / 1.280 are still well inside the bound $B_{1,p}/U_p = 1.404 / 1.804$ of §A.3.

Monte Carlo. The literal two-pass (algorithm 0) and three-pass (A and C: masked smoother, leverage from the same factor, enlargement, inpainting, unmasked smoother, for C the subtraction of S_0u) procedures were simulated on the worst masks, nothing using the identities or the variance formulas; errors are the scatter of independent workers. Algorithm 0, 10^5 realizations per mask, both box32 and w64 on each of four masks (the scan-best periodic (48,192) and (64,128), clamped, and the two annealed worst masks): the window means agree with the exact values to 10^{-3} — e.g. box32 on its annealed mask 0.91469 ± 0.00067 against 0.91417, w64 on its 0.87709 ± 0.00093 against 0.87754 — the eight differences being -0.5 to $+1.8\sigma$ (mean pull $+0.6$); pointwise over the 1024 positions, pull r.m.s. 0.97–1.13 and maximum $|\text{pull}|$ 2.9–3.8 (e.g. box32 on an island 512 samples into the window: 0.5997 against 0.6008). A and C at $\mu = 2$, 30000 realizations per mask, six masks (the four annealed worst masks and the scan-best (64,97) mask), both variants and both kernels on each, 24 comparisons: e.g. A on its box32 worst mask 0.93251 ± 0.00110 against 0.93124, on its w64 worst mask 0.90190 ± 0.00149 against 0.90001; all within $0.6\text{--}1.4\sigma$ (all positive, the six masks sharing their realizations), pointwise pull r.m.s. 0.84–1.07, maximum 3.5. The C worst masks and the scan-best mask are the same structure (nothing enlarged in the window) and give the same numbers.

What is verified and what is best found. Exact, or confirmed by Monte Carlo: the evaluators and the rule (the reference numbers above, the dense algebra, $V' = U$ for $m \equiv 1$, and, directly from (A.5), $V'_C = U$ and $V'_A = U + \|R_m S_0 k\|^2 \geq U$ when nothing is kept, so a wholly enlarged window contributes nothing to $\tilde{X}_{10}$); the scan values (exact for every mask on the grid, phase maximized exactly); the window averages of the ten Monte Carlo masks. Best found, not certified: X_{hi} (algorithm 0: 4 seeds to 6×10^{-4} , reproducing the 7-seed optima; A, and C at $W \leq 16$: 5 seeds to ≤ 0.003 ; C at w48 and w64: seeds spread 0.01–0.04, so $X_{\text{hi}}(C)$ may be somewhat larger); $\tilde{X}_{10}$ from annealing (algorithm 0: 6 seeds within 0.002–0.006 at 4000 steps, all converging to the scan's island structure, so the true value is probably within ~ 0.005 of the one given, but a different structure cannot be excluded; A and C: 2 seeds $\times$ 1200 steps agreeing to ≤ 0.005 , the annealer barely improving the scan, the $\mu = 4$ box32 value 0.083 below algorithm 0's 0.086 for lack of steps, and the $\mu \leq 1.5$ A values, where partial enlargement matters, the least certain). The island floors quoted as X_{10} for the kernels outside Table 2 are analytic candidates, not annealed.

alg.	μ	p0	box8	box16	box32	w48	w64
X_{hi} (annealing, best found)							
0	∞	0.0003	0.0054	0.0114	0.0247	0.0349	0.0504
A	1.2	0.0003	0.0045	0.0095	0.0205	0.0288	0.0418
A	1.5	0.0003	0.0048	0.0102	0.0218	0.0308	0.0450
A	2	0.0003	0.0050	0.0106	0.0232	0.0329	0.0478
A	3	0.0003	0.0053	0.0111	0.0244	0.0343	0.0498
A	4	0.0003	0.0054	0.0113	0.0246	0.0348	0.0502
C	1.2	0.0034	0.0323	0.0657	0.1361	0.1928	0.2797
C	1.5	0.0020	0.0252	0.0561	0.1107	0.1691	0.2179
C	2	0.0009	0.0187	0.0407	0.0826	0.1114	0.1799
C	3	0.0003	0.0130	0.0259	0.0573	0.0796	0.1085
C	4	0.0003	0.0101	0.0196	0.0430	0.0663	0.0866
$\tilde{X}_{lo}, T_c = 1024$ (annealing, best found)							
0	∞	0.0040	0.0281	0.0505	0.0858	0.1046	0.1225
A	1.2	0.0015	0.0111	0.0206	0.0349	0.0416	0.0474
A	1.5	0.0021	0.0152	0.0277	0.0473	0.0574	0.0677
A	2	0.0028	0.0207	0.0383	0.0688	0.0844	0.1000
A	3	0.0033	0.0241	0.0445	0.0774	0.0954	0.1181
A	4	0.0035	0.0249	0.0466	0.0826	0.1046	0.1225
C	1.2	0.0011	0.0071	0.0118	0.0133	0.0116	0.0091
C	1.5	0.0020	0.0141	0.0255	0.0440	0.0525	0.0596
C	2	0.0028	0.0204	0.0380	0.0681	0.0834	0.0984
C	3	0.0033	0.0241	0.0436	0.0765	0.0945	0.1171
C	4	0.0034	0.0249	0.0486	0.0831	0.1047	0.1225
X_{lo} , algorithm 0 (pointwise)		0.996	0.967	0.934	0.871	0.683	0.705

Table 7: The figures of merit of §1.4 at $\tau = 64$ for algorithm 0 ($\mu = \infty$) and for algorithms A and C at five thresholds, for six kernels; Table 3 in §1.4 shows box32 and w64 side by side with the pointwise X_{lo} . Every entry is the best found by simulated annealing (this appendix): X_{hi} by pointwise annealing, $\tilde{X}_{lo}$ by annealing on the window average seeded with the exact scan over periodic masks, which is within 0.012 below the values shown. At $\mu = 4$ the worst windows contain nothing enlarged, so the A and C rows there are algorithm 0’s values up to annealing noise (box32: 0.083 against 0.086 for lack of steps). The last row is the pointwise X_{lo} of algorithm 0 (the unbounded island floors of §A.4; Table 2 has A and C), for comparison with $\tilde{X}_{lo}$.